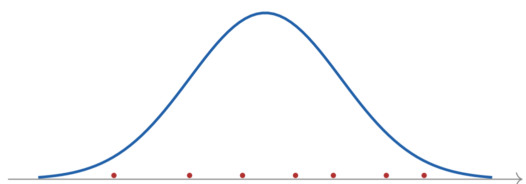
Probability & Statistics

Lecture Notes with Worked Examples

A One-Semester Course



A unified treatment of exploratory data analysis, probability, statistical inference, maximum likelihood, regression, and goodness-of-fit testing.

Course notes and integrated problem set

About These Notes

These notes assemble the material of a one-semester introductory course in probability and statistics into a single self-contained volume. The aim is to present the theory and a large stock of fully worked examples *side by side*, so that every idea is immediately illustrated by concrete computation.

The book is organized into three movements. The first (Chapters 1–6) develops the language of probability: sets and counting, conditional probability and Bayes' rule, discrete and continuous random variables, joint distributions, and the classical limit theorems. The second (Chapter 7) turns to data itself through exploratory data analysis. The third (Chapters 8–12) builds the machinery of statistical inference: point estimation, confidence intervals, maximum likelihood, least-squares regression, and hypothesis testing culminating in goodness-of-fit tests.

Every worked example is woven into the section whose ideas it exercises. A problem about joint distributions appears beside the discussion of joint distributions; a maximum-likelihood computation appears beside the maximum likelihood principle. Where the original course used the statistical software R for computation, the corresponding numerical answers are simply stated, since the emphasis here is on the underlying reasoning rather than on software commands.

A short note on figures: the plots in these notes were regenerated from the underlying datasets (the black cherry tree volumes, the town mortality and calcium measurements, and the small numerical datasets that appear in the examples). The Old Faithful illustrations use a dataset constructed to match the summary statistics quoted in the text and are meant to be illustrative.

Contents

CHAPTER 1

Mathematical Prerequisites

Probability and statistics lean heavily on a handful of tools from calculus. Expectations of continuous random variables are integrals; the maximum likelihood method finds estimates by differentiating and setting the derivative to zero; tail probabilities are computed by integrating density functions. Before we begin, this short chapter revisits the differentiation and integration techniques that recur throughout the book. A reader fluent in single-variable calculus may skip directly to Chapter 2 and refer back as needed.

1.1 Differentiation

The two rules we use most often are the *product rule*

$$\frac{d}{dx}\big(u(x)\,v(x)\big) = u'(x)\,v(x) + u(x)\,v'(x)$$

and the *chain rule*

$$\frac{d}{dx}\,f\big(g(x)\big) = f'\big(g(x)\big)\,g'(x).$$

Together with the quotient rule and the elementary derivatives of e^x, $\ln x$, and the trigonometric functions, these handle essentially every derivative that arises in this course.

Example 1.1

Differentiate $f(x) = x^2 \ln x$.

Solution. The function is a product, so the product rule with $u = x^2$ and $v = \ln x$ gives

$$f'(x) = (2x)\ln x + x^2 \cdot \frac{1}{x} = 2x\ln x + x.$$

Example 1.2

Let $f(x) = xe^{-x}$. Find $f'(x)$ and $f''(x)$.

Solution. Applying the product rule once,

$$f'(x) = e^{-x} + x(-e^{-x}) = e^{-x} - xe^{-x}.$$

Differentiating again, and reusing the result just obtained for the second term,

$$f''(x) = -e^{-x} - \left(e^{-x} - xe^{-x}\right) = -2e^{-x} + xe^{-x}.$$

This particular function reappears when we study the gamma-type densities and integration by parts.

Example 1.3

Differentiate $y = e^{-(x-1)^2/2}$.

Solution. This is a composition, so we peel it apart with the chain rule. The outer function is e^u with $u = -\frac{1}{2}(x-1)^2$, and $u' = -(x-1)$. Hence

$$\frac{dy}{dx} = e^{-(x-1)^2/2} \cdot \big(-(x-1)\big) = -(x-1)\, e^{-(x-1)^2/2}.$$

The exponent here is precisely the kernel of the standard normal density, which is why this derivative shows up when we maximize normal likelihoods.

1.2 Integration

The reverse operation, integration, computes areas and—crucially for us—the expectations and probabilities of continuous random variables. We use basic antiderivatives, linearity, the substitution rule, and integration by parts.

1.2.1 Antiderivatives and definite integrals

Example 1.4

Evaluate $\displaystyle\int \Big(x^3 + e^x\Big)\, dx$ and $\displaystyle\int \Big(2\theta + \tfrac{1}{\theta}\Big)\, d\theta$.

Solution. Integrating term by term,

$$\int \Big(x^3 + e^x\Big)\, dx = \frac{x^4}{4} + e^x + C, \qquad \int \Big(2^\theta + \tfrac{1}{\theta}\Big)\, d\theta = \frac{2^\theta}{\ln 2} + \ln|\theta| + C.$$

(The exponential 2^θ has antiderivative $2^\theta/\ln 2$ because $\frac{d}{d\theta}2^\theta = 2^\theta \ln 2$.)

Example 1.5

Evaluate the definite integral $\displaystyle\int_0^{\pi/2} -2\cos\theta\, d\theta$.

Solution. An antiderivative of $-2\cos\theta$ is $-2\sin\theta$, so

$$\int_0^{\pi/2} -2\cos\theta\, d\theta = \big[-2\sin\theta\big]_0^{\pi/2} = -2\sin(\tfrac{\pi}{2}) + 2\sin 0 = -2.$$

1.2.2 Integration by parts

The integration-by-parts formula,

$$\int u\, dv = uv - \int v\, du,$$

is indispensable for computing expectations of the form $\int x f(x)\,dx$ when f involves an exponential. We will use it repeatedly—for instance to show that an $\mathrm{Exp}(\lambda)$ random variable has mean $1/\lambda$, and that the square of a standard normal random variable has expectation 1.

Example 1.6

Use integration by parts to evaluate $\displaystyle\int_0^\infty x\,\lambda e^{-\lambda x}\,dx$ for $\lambda > 0$.

Solution. Take $u = x$ and $dv = \lambda e^{-\lambda x}\,dx$, so that $du = dx$ and $v = -e^{-\lambda x}$. Then

$$\int_0^\infty x\,\lambda e^{-\lambda x}\,dx = \Big[-x e^{-\lambda x}\Big]_0^\infty + \int_0^\infty e^{-\lambda x}\,dx = 0 + \left[-\frac{1}{\lambda}e^{-\lambda x}\right]_0^\infty = \frac{1}{\lambda}.$$

This is exactly the mean of an exponential random variable with rate λ, a fact we will meet again in Chapter 4.

With these tools in hand we are ready to build the theory of probability.

CHAPTER 2

The Language of Probability

Probability theory is a precise language for reasoning about uncertainty. Its basic objects are *experiments* whose outcomes cannot be predicted with certainty, and its goal is to assign numbers—probabilities—to the various things that might happen. This chapter introduces that language: sample spaces and events, the rules probabilities must obey, counting, and the profoundly useful ideas of conditional probability, independence, and Bayes' theorem.

2.1 Sample Spaces and Events

The set of all possible outcomes of an experiment is the *sample space*, denoted Ω. An *event* is a subset of Ω: a collection of outcomes we might be interested in. Because events are sets, the operations of set theory—union ($\cup$, "or"), intersection ($\cap$, "and"), and complement (A^c, "not")—are exactly the operations we use to combine and negate events.

Example 2.1

A coin is tossed three times, recording heads (H) or tails (T) each time, so

$$\Omega = \{HHH,\ THH,\ HTH,\ HHT,\ TTH,\ THT,\ HTT,\ TTT\}.$$

Consider the events $A =$"exactly two tails," $B =$"at least two tails," $C =$"no head precedes the first tail" (equivalently, tails never appears after a head), and $D =$"the first toss is tails." Express A, B, C, D as sets, and then find A^c, $A \cup (C \cap D)$, and $A \cap D^c$.

Solution. Reading off the outcomes,

$$A = \{TTH, THT, HTT\}, \quad B = \{TTH, THT, HTT, TTT\},$$

$$C = \{HHH, HHT, HTT, TTT\}, \quad D = \{THH, TTH, THT, TTT\}.$$

The complement of A contains every outcome that is *not* in A:

$$A^c = \{HHH, THH, HTH, HHT, TTT\}.$$

Since $C \cap D = \{TTT\}$ (the only outcome in both), we have $A \cup (C \cap D) = \{TTT, TTH, THT, HTT\}$. Finally $D^c = \{HHH, HTH, HHT, HTT\}$, and the only outcome common to A and D^c is HTT, so $A \cap D^c = \{HTT\}$.

Sometimes it is the *structure* of the experiment, rather than an explicit list, that tells us what the sample space should be.

Example 2.2

A coin with $\mathbb{P}(\text{heads}) = p$ is tossed repeatedly until heads appears for the *second* time. What is a natural sample space, and what is the probability that exactly five tosses are required?

Solution. The quantity of interest is the toss on which the second head lands; this is at least 2, so $\Omega = \{2, 3, 4, \dots\}$. For the second head to occur on toss 5, exactly one of the first four tosses is a head and the fifth is a head. There are four positions for that early head, each sequence having probability $p^2(1-p)^3$, so the answer is

$$4\,p^2(1-p)^3.$$

2.2 The Rules of Probability

A probability assignment must satisfy a few axioms: every event has probability between 0 and 1, the whole sample space has probability 1, and the probability of a union of *disjoint* events is the sum of their probabilities. From these follow the rules we use constantly. The most important is the *addition rule* (inclusion–exclusion for two events):

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

The subtracted term corrects for the overlap, which would otherwise be counted twice.

Example 2.3

Suppose $\mathbb{P}(A) = \frac{2}{3}$, $\mathbb{P}(B) = \frac{1}{6}$, and $\mathbb{P}(A \cap B) = \frac{1}{9}$. Find $\mathbb{P}(A \cup B)$.

Solution. By the addition rule,

$$\mathbb{P}(A \cup B) = \tfrac{2}{3} + \tfrac{1}{6} - \tfrac{1}{9} = \tfrac{12}{18} + \tfrac{3}{18} - \tfrac{2}{18} = \tfrac{13}{18}.$$

Example 2.4

For events with $\mathbb{P}(A) = 0.4$, $\mathbb{P}(B) = 0.5$, and $\mathbb{P}(A \cap B) = 0.1$, find the probability that *exactly one* of A, B occurs.

Solution. "A or B but not both" is the event $A \cup B$ with the overlap removed. Since $\mathbb{P}(A \cup B) = 0.4 + 0.5 - 0.1 = 0.8$, the probability of exactly one is

$$\mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.8 - 0.1 = 0.7.$$

The addition rule also lets us *bound* probabilities even when we do not know the overlap exactly.

Example 2.5

Two disasters D_1 and D_2 are each rare: $\mathbb{P}(D_1) \le 10^{-6}$ and $\mathbb{P}(D_2) \le 10^{-6}$. What can be said about the probability that at least one occurs, and that both occur?

Solution. Because probabilities are nonnegative, dropping the (subtracted) intersection term can only increase the bound:

$$\mathbb{P}(D_1 \cup D_2) = \mathbb{P}(D_1) + \mathbb{P}(D_2) - \mathbb{P}(D_1 \cap D_2) \le 10^{-6} + 10^{-6} = 2 \times 10^{-6}.$$

For the intersection, note $D_1 \cap D_2 \subseteq D_1$, so monotonicity gives $\mathbb{P}(D_1 \cap D_2) \leq \mathbb{P}(D_1) \leq 10^{-6}$.

2.3 Counting

When every outcome of a finite experiment is equally likely, computing a probability reduces to counting: $\mathbb{P}(A)$ is the number of outcomes in A divided by the total number of outcomes. The binomial coefficient $\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$ counts the number of ways to choose k items from n without regard to order.

Example 2.6

A system has five components, each either working (1) or failed (0), so an outcome is a vector $(x_1, \ldots, x_5)$. In how many ways can it happen that exactly two components work and three have failed?

Solution. We must choose which 2 of the 5 positions hold a 1; the rest are 0. The number of such choices is $\binom{5}{2} = 10$. (Explicitly, the ten outcomes range from $(1, 1, 0, 0, 0)$ through $(0, 0, 0, 1, 1)$.)

2.4 Conditional Probability and Independence

The *conditional probability* of A given B rescales probabilities to the world in which B is known to have occurred:

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \qquad \mathbb{P}(B) \neq 0.$$

Two events are *independent* if knowing one tells us nothing about the other—formally, if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, equivalently $\mathbb{P}(A \mid B) = \mathbb{P}(A)$.

Example 2.7

A fair die is thrown twice. Let A be "the two throws sum to 4" and B be "at least one throw is a 3." Compute $\mathbb{P}(A \mid B)$, and decide whether A and B are independent.

Solution. Event B consists of the eleven equally likely outcomes containing a 3. Of these, only $(1, 3)$ and $(3, 1)$ sum to 4, so $\mathbb{P}(A \mid B) = \frac{2}{11}$. For independence, compare: $A = \{(1, 3), (3, 1), (2, 2)\}$ has $\mathbb{P}(A) = \frac{3}{36} = \frac{1}{12}$, which is *not* equal to $\mathbb{P}(A \mid B) = \frac{2}{11}$. The events are therefore dependent.

Independence makes "and" probabilities multiply, which is what lets us model repeated trials.

Example 2.8

Two laboratory experiments are run each day in separate rooms, independently, until at least one succeeds; each succeeds with probability p. What is the probability that the first success occurs on day n?

Solution. On any single day the probability that *at least one* room succeeds is, by the addition rule and independence,

$$q := \mathbb{P}(\text{success on a given day}) = p + p - p^2 = 2p - p^2.$$

Reaching day n with the first success requires $n-1$ failed days followed by a successful one. Since days are independent,

$$\mathbb{P}(\text{first success on day } n) = (1-q)^{n-1}q = \big(1-(2p-p^2)\big)^{n-1}(2p-p^2).$$

2.5 The Law of Total Probability and Bayes' Theorem

Often we know the probability of an event *conditional* on each of several scenarios, and want the overall probability. The *law of total probability* stitches the scenarios together: if B and B^c partition the sample space,

$$\mathbb{P}(A) = \mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c).$$

Reversing the direction of conditioning is the job of *Bayes' theorem*:

$$\mathbb{P}(B \mid A) = \frac{\mathbb{P}(A \mid B)\,\mathbb{P}(B)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A \mid B)\mathbb{P}(B)}{\mathbb{P}(A \mid B)\mathbb{P}(B) + \mathbb{P}(A \mid B^c)\mathbb{P}(B^c)}.$$

This is the workhorse of medical testing, reliability, and inference. The following examples show its range.

Example 2.9 (Diagnostic test for a rare condition)
A cow is tested for BSE with sensitivity $\mathbb{P}(T \mid B) = 0.70$ and false-positive rate $\mathbb{P}(T \mid B^c) = 0.10$, where B is "the cow is infected" and T is "the test is positive." The prevalence is $\mathbb{P}(B) = 1.3 \times 10^{-5}$. Find $\mathbb{P}(B \mid T)$.

Solution. Bayes' theorem with the law of total probability in the denominator gives

$$\mathbb{P}(B \mid T) = \frac{0.70\,(1.3 \times 10^{-5})}{0.70\,(1.3 \times 10^{-5}) + 0.10\,(1 - 1.3 \times 10^{-5})} \approx 9 \times 10^{-5}.$$

Even after a positive test the cow is almost certainly healthy: when a condition is extremely rare, false positives vastly outnumber true positives. This counterintuitive phenomenon is the *base-rate* effect.

Example 2.10 (Solving for the test quality)
A breath analyzer satisfies $\mathbb{P}(A \mid B) = \mathbb{P}(A^c \mid B^c) = p$, where A is "the analyzer reports over the limit" and B is "the driver is truly over the limit." On a Saturday night, 5% of drivers are over the limit. How large must p be so that $\mathbb{P}(B \mid A) = 0.9$?

Solution. By Bayes' theorem,

$$0.9 = \mathbb{P}(B \mid A) = \frac{p(0.05)}{p(0.05) + (1-p)(0.95)}.$$

Clearing the denominator, $0.9\big(0.05p + 0.95(1-p)\big) = 0.05p$, which rearranges to $\big(\frac{0.05}{0.9} + 0.9\big)p = 0.95$, giving $p \approx 0.994$. A very accurate device is needed before a positive reading is convincing, again because the underlying event is uncommon.

Example 2.11 (Two independent tests)

A disease has prevalence $\mathbb{P}(D) = 0.01$. A test satisfies $\mathbb{P}(T \mid D) = 0.98$ and $\mathbb{P}(T^c \mid D^c) = 0.95$. (a) Given one positive test, find $\mathbb{P}(D \mid T)$. (b) A second, independent test also comes back positive; now find $\mathbb{P}(D \mid T_1 \cap T_2)$.

Solution. (a) Bayes' theorem yields

$$\mathbb{P}(D \mid T) = \frac{0.98(0.01)}{0.98(0.01) + 0.05(0.99)} = \frac{0.0098}{0.0593} \approx 0.166.$$

(b) The two tests are independent *conditional on disease status*, so

$$\begin{aligned}\mathbb{P}(D \mid T_1 \cap T_2) &= \frac{\mathbb{P}(T_1 \mid D)\mathbb{P}(T_2 \mid D)\mathbb{P}(D)}{\mathbb{P}(T_1 \mid D)\mathbb{P}(T_2 \mid D)\mathbb{P}(D) + \mathbb{P}(T_1 \mid D^c)\mathbb{P}(T_2 \mid D^c)\mathbb{P}(D^c)} \\ &= \frac{(0.98)^2(0.01)}{(0.98)^2(0.01) + (0.05)^2(0.99)} \approx 0.795.\end{aligned}$$

A subtle point. It is tempting to write $\mathbb{P}(T_1 \cap T_2) = \mathbb{P}(T_1)\mathbb{P}(T_2)$, but that is wrong: the tests are *not* unconditionally independent. They are independent only once disease status is fixed. Indeed a first positive raises the chance of disease, which in turn raises the chance the second test is positive. This distinction—between independence and *conditional* independence—is one of the most important in all of probability.

Example 2.12 (A multiple-choice exam)

On each question a student either knows the answer (then answers correctly with probability 1) or guesses among the options (then correct with probability $\frac{1}{4}$). The student knows each answer with probability 0.6. Given that a question is answered correctly, what is the probability the student actually knew it?

Solution. Let $B =$"knows the answer," $A =$"answers correctly." Then $\mathbb{P}(B) = 0.6$, $\mathbb{P}(A \mid B) = 1$, $\mathbb{P}(A \mid B^c) = \frac{1}{4}$, and Bayes' theorem gives

$$\mathbb{P}(B \mid A) = \frac{1 \cdot 0.6}{1 \cdot 0.6 + \frac{1}{4} \cdot 0.4} = \frac{0.6}{0.7} = \frac{6}{7}.$$

Example 2.13 (The neighbor and the plant)

While you are away, a neighbor waters a sickly plant. Without water it dies with probability 0.8; with water it dies with probability 0.15. You are 90% sure the neighbor remembers. (a) Find the probability the plant is alive on your return. (b) If it is dead, find the probability the neighbor forgot.

Solution. Let $A =$"alive" and $R =$"remembers to water." By the law of total probability,

$$\mathbb{P}(A) = \mathbb{P}(A \mid R)\mathbb{P}(R) + \mathbb{P}(A \mid R^c)\mathbb{P}(R^c) = (0.85)(0.9) + (0.2)(0.1) = \tfrac{157}{200}.$$

For part (b), Bayes' theorem with A^c gives

$$\mathbb{P}(R^c \mid A^c) = \frac{\mathbb{P}(A^c \mid R^c)\mathbb{P}(R^c)}{\mathbb{P}(A^c)} = \frac{(0.8)(0.1)}{1 - \frac{157}{200}} = \frac{0.08}{43/200} = \frac{16}{43}.$$

Example 2.14 (Inferring an infestation from a union)

Grapefruit is grown in two regions, R_1 and R_2. Let A and B be the events that R_1 and R_2 respectively are infested with a parasite, with $\mathbb{P}(A) = \frac{3}{4}$, $\mathbb{P}(B) = \frac{2}{5}$, and $\mathbb{P}(A \cup B) = \frac{4}{5}$. If a shipment from R_1 is found infested, what is the probability R_2 is infested too?

Solution. We want $\mathbb{P}(B \mid A) = \mathbb{P}(A \cap B)/\mathbb{P}(A)$. The addition rule recovers the intersection: $\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) = \frac{3}{4} + \frac{2}{5} - \frac{4}{5} = \frac{7}{20}$. Hence $\mathbb{P}(B \mid A) = \frac{7/20}{3/4} = \frac{7}{15}$.

2.5.1 Bayes' theorem with geometric structure: the three-card problem

Example 2.15 (Three cards)

Three cards are placed in a hat: one green on both sides, one red on both sides, and one green on one side and red on the other. You draw a card and look at one side, both chosen at random. The side you see is green. What is the probability the other side is also green?

Solution. Let G_1 be "the visible side is green" and G_2 be "the hidden side is green." The only way *both* sides are green is the green–green card, so $\mathbb{P}(G_1 \cap G_2) = \frac{1}{3}$. By symmetry $\mathbb{P}(G_1) = \frac{1}{2}$ (half of all six card-faces are green). Therefore

$$\mathbb{P}(G_2 \mid G_1) = \frac{\mathbb{P}(G_1 \cap G_2)}{\mathbb{P}(G_1)} = \frac{1/3}{1/2} = \frac{2}{3}.$$

The intuitive guess of $\frac{1}{2}$ is wrong: seeing a green face makes the double-green card more likely than the mixed card, because the double-green card has *two* ways to show you a green face.

2.5.2 The Monty Hall problem

Example 2.16 (Monty Hall)

A car sits behind one of three doors; goats sit behind the other two. You pick a door, and the host—who knows where the car is—opens a different door to reveal a goat. Should you switch? Label the door hiding the car a and the others b, c, and model the experiment by the product space $\Omega = \{a, b, c\} \times \{a, b, c\}$, where the first coordinate is your choice and the second is the door the host opens.

Solution. The host never opens door a (the car is there), so any pair with second coordinate a has probability 0. If you pick a (probability $\frac{1}{3}$), the host chooses b or c equally, giving $\mathbb{P}((a, b)) = \mathbb{P}((a, c)) = \frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$. If you pick b, the host must open c, so $\mathbb{P}((b, c)) = \frac{1}{3}$; similarly $\mathbb{P}((c, b)) = \frac{1}{3}$.
A *no-switch* player wins exactly when the original pick is a, an event of probability $\frac{1}{6} + \frac{1}{6} = \frac{1}{3}$.
A *switching* player wins exactly when the original pick was b or c, namely on $\{(b, c), (c, b)\}$, of probability $\frac{1}{3} + \frac{1}{3} = \frac{2}{3}$. Switching doubles your chances.

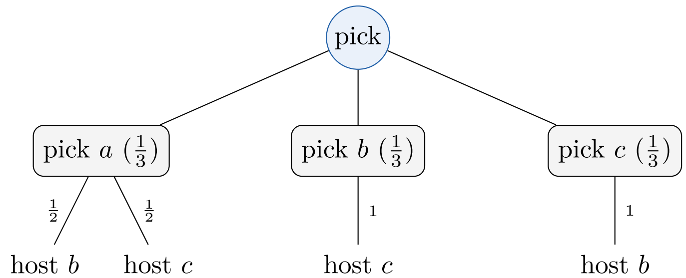


Figure 2.1. The Monty Hall experiment. Staying wins only on the left branch (probability $\frac{1}{3}$); switching wins on the other two (probability $\frac{2}{3}$).

CHAPTER 3

Discrete Random Variables

A *random variable* is a numerical summary of the outcome of an experiment—a function that assigns a number to each point of the sample space. A random variable is *discrete* when it takes values in a finite or countable set, such as the number of heads in ten tosses or the number of trials until a success. This chapter develops the standard discrete models and the two numbers—expectation and variance—that summarize them.

3.1 Probability Mass and Distribution Functions

The *probability mass function* (pmf) of a discrete random variable X is $p_X(a) = \mathbb{P}(X = a)$. It is nonnegative and sums to one over the possible values. The *cumulative distribution function* (cdf) accumulates the mass, $F_X(a) = \mathbb{P}(X \le a)$, and is a nondecreasing step function that rises from 0 to 1.

3.2 Expectation and Variance

The *expectation* (or mean) of X is the probability-weighted average of its values,

$$\mathbb{E}[X] = \sum_a a\, p_X(a),$$

and the *variance* measures spread about the mean,

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mathbb{E}[X])^2\big] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The second form on the right is almost always the easier one to compute. Expectation is *linear*: $\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$ for any random variables and constants, whether or not X and Y are independent.

Example 3.1 (A fair four-sided die)

A four-sided die shows $X \in \{1, 2, 3, 4\}$, each with probability $\frac{1}{4}$. Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution. The mean is

$$\mathbb{E}[X] = \tfrac{1}{4}(1 + 2 + 3 + 4) = \tfrac{10}{4} = \tfrac{5}{2}.$$

For the variance, first $\mathbb{E}[X^2] = \frac{1}{4}(1 + 4 + 9 + 16) = \frac{30}{4}$, so

$$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \tfrac{30}{4} - \tfrac{25}{4} = \tfrac{5}{4}.$$

3.2.1 Functions of a discrete random variable

To find the distribution of $Y = g(X)$, we collect the probability of each value of X onto the corresponding value of Y.

Example 3.2

Let X take the values $80, 90, 100, 110, 120$, each with probability 0.2, so $\mathbb{E}[X] = 100$. Find the distribution of $Y = |X - 100|$.

Solution. The map $x \mapsto |x - 100|$ sends $80, 120 \mapsto 20$, sends $90, 110 \mapsto 10$, and sends $100 \mapsto 0$. Collecting probabilities,

$$\mathbb{P}(Y = 0) = \tfrac{1}{5}, \qquad \mathbb{P}(Y = 10) = \tfrac{2}{5}, \qquad \mathbb{P}(Y = 20) = \tfrac{2}{5}.$$

Example 3.3

Let X have pmf $p(-1) = \frac{1}{4},\ p(0) = \frac{1}{8},\ p(1) = \frac{1}{8},\ p(2) = \frac{1}{2}$. Find the pmf of $Y = X^2$.

Solution. Now $x \mapsto x^2$ sends -1 and 1 both to 1, sends $0 \mapsto 0$, and sends $2 \mapsto 4$. Adding the masses that collide,

$$\mathbb{P}(Y = 0) = \tfrac{1}{8}, \qquad \mathbb{P}(Y = 1) = \tfrac{1}{4} + \tfrac{1}{8} = \tfrac{3}{8}, \qquad \mathbb{P}(Y = 4) = \tfrac{1}{2}.$$

3.3 The Bernoulli and Binomial Distributions

A *Bernoulli* trial has two outcomes, "success" (coded 1) with probability p and "failure" (coded 0) with probability $1 - p$; we write $X \sim \mathrm{Ber}(p)$. The number of successes in n independent Bernoulli trials, all with the same p, is *binomial*, $X \sim \mathrm{Bin}(n, p)$, with pmf

$$\mathbb{P}(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Its mean and variance are $\mathbb{E}[X] = np$ and $\mathrm{Var}(X) = np(1 - p)$.

Example 3.4 (Counting defectives)

A shop receives four lamps, each defective with probability $\frac{1}{3}$, independently. Let X be the number of defective lamps. What kind of distribution does X have, and what is $\mathbb{P}(X = 2)$?

Solution. Here $X \sim \mathrm{Bin}(4, \frac{1}{3})$. Then

$$\mathbb{P}(X = 2) = \binom{4}{2} \left(\tfrac{1}{3}\right)^2 \left(\tfrac{2}{3}\right)^2 = 6 \cdot \tfrac{1}{9} \cdot \tfrac{4}{9} = \tfrac{24}{81} = \tfrac{8}{27}.$$

Example 3.5 (An eight-engine bomber)

A bomber has eight engines, each critically damaged on a mission with probability 0.10, independently. It returns safely if at least six engines keep working. What is the probability of a safe return?

Solution. Let X be the number of *working* engines, so $X \sim \mathrm{Bin}(8, 0.90)$ (working is the success). A safe return is $X \geq 6$:

$$\mathbb{P}(X \geq 6) = \sum_{k=6}^{8} \binom{8}{k} (0.9)^k (0.1)^{8-k} = 28(0.9)^6(0.1)^2 + 8(0.9)^7(0.1) + (0.9)^8 \approx 0.96.$$

Example 3.6 (Guessing on a multiple-choice test)

A test has 20 questions, each with four options. You must answer at least 65% correctly to pass. If you guess every question, what are your chances?

Solution. Let $X \sim \mathrm{Bin}(20, \frac{1}{4})$ be the number correct. Since 65% of 20 is 13, passing means $X \geq 13$:

$$\mathbb{P}(X \geq 13) = 1 - \sum_{k=0}^{12} \binom{20}{k} (\tfrac{1}{4})^k (\tfrac{3}{4})^{20-k} \approx 0.00018,$$

about 0.02%. Guessing your way to a pass is, for practical purposes, hopeless.

Example 3.7 (An election, and the value of a larger sample)

In a town, 55% of voters prefer Robby; a simple majority wins. (a) If 41 people vote, what is the probability Robby wins? (b) If 401 vote?

Solution. (a) Let $X \sim \mathrm{Bin}(41, 0.55)$; Robby wins if $X \geq 21$, and $\mathbb{P}(X \geq 21) \approx 0.741$. (b) With $Y \sim \mathrm{Bin}(401, 0.55)$ we need $Y \geq 201$, and $\mathbb{P}(Y \geq 201) \approx 0.978$. Even though the underlying preference is unchanged, the larger electorate makes the majority outcome far more reliable—an early glimpse of the law of large numbers (Chapter 6).

3.4 The Geometric Distribution

The number of independent $\mathrm{Ber}(p)$ trials needed to obtain the *first* success is *geometric*, $X \sim \mathrm{Geo}(p)$, with

$$\mathbb{P}(X = k) = (1-p)^{k-1} p, \qquad k = 1, 2, 3, \ldots$$

Its mean is $\mathbb{E}[X] = 1/p$. The geometric distribution is the discrete analogue of the memoryless exponential distribution we will meet in Chapter 4.

Example 3.8 (Two lotteries)

Each month you play two lotteries, winning a large prize with probabilities p_1 and p_2 respectively, and you stop as soon as you win in either. Let M be the number of months you play. What is the distribution of M?

Solution. In a single month the chance of *some* win is, by inclusion and independence, $q = p_1 + p_2 - p_1 p_2$. Stopping in month n requires $n-1$ winless months followed by a win, so $\mathbb{P}(M = n) = (1-q)^{n-1} q$. Thus $M \sim \mathrm{Geo}(p_1 + p_2 - p_1 p_2)$.

Example 3.9 (Cycles to pregnancy)

The number of menstrual cycles until pregnancy is modeled as $X \sim \mathrm{Geo}(\frac{1}{3})$. (a) What is the most likely value of X? (b) Find the probability of success within five cycles.

Solution. (a) Because $\mathbb{P}(X = k) = p(1-p)^{k-1}$ is decreasing in k (as $0 < 1 - p < 1$), the most likely value is $k = 1$. (b) Summing the pmf,

$$\mathbb{P}(X \leq 5) = \sum_{k=1}^{5} (1-p)^{k-1} p \approx 0.868,$$

roughly an 87% chance within five cycles. (We return to this dataset when we estimate p from real data in Chapters 9 and 10.)

Example 3.10 (A concert ticket and a fair-seeming rule)

Only one ticket remains, and a seller tosses a coin (heads with probability p) until heads appears. If the number of tosses is odd your friend gets the ticket; if even, you do. Is this fair?

Solution. Let $X \sim \mathrm{Geo}(p)$. Summing the odd terms gives a geometric series with ratio $(1-p)^2$:

$$\mathbb{P}(X \text{ odd}) = p \sum_{j \geq 0} (1-p)^{2j} = \frac{p}{1-(1-p)^2} = \frac{1}{2-p},$$

and therefore $\mathbb{P}(X \text{ even}) = \frac{1-p}{2-p}$. Since $0 < p < 1$,

$$\mathbb{P}(X \text{ even}) = \frac{1-p}{2-p} < \frac{1}{2-p} = \mathbb{P}(X \text{ odd}).$$

The rule favors your friend, so you should decline. (When $p = \frac{1}{2}$ the odds are $\frac{2}{3}$ to $\frac{1}{3}$.)

Example 3.11 (Estimating a population by recapture)

A box holds an unknown number N of identical bolts. Mark one bolt; then draw bolts at random one at a time. (a) If you sample *with* replacement until the marked bolt reappears, what is the distribution of the number X of draws? (b) If you sample *without* replacement, show the number of draws Y is uniform on $\{1, 2, \ldots, N\}$.

Solution. (a) Each draw is the marked bolt with probability $1/N$ independently, so $X \sim \mathrm{Geo}(1/N)$. (b) Without replacement, the marked bolt is equally likely to occupy any position in the random order of removal, so $\mathbb{P}(Y = k) = 1/N$ for each $k = 1, \ldots, N$. One checks this directly: $\mathbb{P}(Y = 1) = 1/N$, and by the law of total probability $\mathbb{P}(Y = 2) = \frac{1}{N-1} \cdot \frac{N-1}{N} = \frac{1}{N}$, and so on. This is the *discrete uniform* distribution; we estimate N from such data in Chapter 10.

3.5 The Poisson Distribution

The *Poisson* distribution models counts of rare events—arrivals at a server, accidents at an intersection, bomb hits on a grid—when events occur independently at a constant average rate μ.

We write $X \sim \text{Pois}(\mu)$, with

$$\mathbb{P}(X = k) = \frac{\mu^k e^{-\mu}}{k!}, \qquad k = 0, 1, 2, \ldots$$

and $\mathbb{E}[X] = \text{Var}(X) = \mu$.

Example 3.12 (Does the cold drug help?)

A person catches Pois(3) colds per year normally. A new drug lowers the mean to Pois(2) for 75% of people and does nothing for the other 25%. A person tries it for a year and catches *no* colds. How likely is it that the drug helped them?

Solution. Let $W =$"no colds" and $D =$"the drug helps." For a Poisson variable, $\mathbb{P}(\text{no colds}) = e^{-\mu}$, so $\mathbb{P}(W \mid D) = e^{-2}$ and $\mathbb{P}(W \mid D^c) = e^{-3}$. By Bayes' theorem,

$$\mathbb{P}(D \mid W) = \frac{e^{-2}(0.75)}{e^{-2}(0.75) + e^{-3}(0.25)} \approx 0.89.$$

An uneventful year is decent—though not conclusive—evidence that the drug worked.

The Poisson model reappears in Chapter 9, where we fit it to the famous data on World War II bomb hits over South London, and in Chapter 12, where we test how well it fits counts of traffic accidents.

CHAPTER 4

Continuous Random Variables

A *continuous* random variable takes values in a continuum—a waiting time, a length, a temperature—and no single value carries positive probability. Instead, probability is spread out as a *density*, and probabilities of intervals are areas under that density. This chapter develops the continuous analogues of the previous chapter's ideas and surveys the uniform, normal, exponential, Pareto, and Weibull families.

4.1 Densities, Distribution Functions, and Expectation

A continuous random variable X has a *probability density function* (pdf) f with $f(x) \geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx = 1$. Probabilities are integrals of the density,

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x)\,dx,$$

and the *cumulative distribution function* is $F(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$. Expectation and variance are the continuous counterparts of the discrete formulas, with sums replaced by integrals:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx, \qquad \mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The *median* m splits the area in half: $\int_{-\infty}^{m} f(x)\,dx = \frac{1}{2}$.

Example 4.1 (Reading probabilities off a cdf)
Suppose $F(x) = x^2$ for $0 \leq x \leq 1$. Find $\mathbb{P}(\frac{1}{2} \leq X \leq \frac{3}{4})$.

Solution. Probabilities of intervals are differences of the cdf:

$$\mathbb{P}(\tfrac{1}{2} \leq X \leq \tfrac{3}{4}) = F(\tfrac{3}{4}) - F(\tfrac{1}{2}) = \tfrac{9}{16} - \tfrac{1}{4} = \tfrac{5}{16}.$$

Example 4.2 (From a piecewise density to its cdf)
Let X have density $f(x) = \frac{1}{4}$ on $0 < x < 1$, $f(x) = \frac{3}{8}$ on $3 < x < 5$, and $f(x) = 0$ otherwise. Find the cdf.

Solution. Integrate the density from the left, accumulating area. On $[0, 1]$ the area grows like $\frac{1}{4}x$, reaching $\frac{1}{4}$ at $x = 1$; nothing accumulates on $[1, 3]$; on $[3, 5]$ the area grows by $\frac{3}{8}(x - 3)$ on top of the $\frac{1}{4}$ already banked. Hence

$$F(x) = \begin{cases} 0, & x \le 0, \\ \frac{1}{4}x, & 0 < x < 1, \\ \frac{1}{4}, & 1 \le x \le 3, \\ \frac{3}{8}x - \frac{7}{8}, & 3 < x < 5, \\ 1, & x \ge 5. \end{cases}$$

The flat stretch on $[1, 3]$ reflects the gap where X places no probability.

Example 4.3 (Mean and variance of a polynomial density)
Let $f(x) = 3x^2$ for $0 \le x \le 1$ and 0 otherwise. Compute $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.
Solution. Directly,

$$\mathbb{E}[X] = \int_0^1 x \cdot 3x^2 \, dx = \left[\tfrac{3}{4}x^4\right]_0^1 = \tfrac{3}{4},$$

and $\mathbb{E}[X^2] = \int_0^1 3x^4 \, dx = \frac{3}{5}$, so

$$\mathrm{Var}(X) = \tfrac{3}{5} - \left(\tfrac{3}{4}\right)^2 = \tfrac{3}{5} - \tfrac{9}{16} = \tfrac{3}{80}.$$

Example 4.4 (An expectation requiring integration by parts)
The lifetime of a vacuum tube has density $f(x) = xe^{-x}$ for $x \ge 0$. Find the expected lifetime.
Solution. We must integrate $\mathbb{E}[X] = \int_0^\infty x^2 e^{-x} \, dx$. Integrating by parts once turns x^2 into $2x$, and a second time turns $2x$ into 2:

$$\int_0^\infty x^2 e^{-x} \, dx = \left[-x^2 e^{-x}\right]_0^\infty + \int_0^\infty 2xe^{-x} \, dx = 0 + 2\int_0^\infty xe^{-x} \, dx = 2.$$

So the expected lifetime is 2 hours.

4.2 The Uniform Distribution

If X is equally likely to fall anywhere in $[\alpha, \beta]$, it is *uniform*, $X \sim \mathrm{Unif}(\alpha, \beta)$, with constant density $f(x) = 1/(\beta - \alpha)$ on $[\alpha, \beta]$.

Example 4.5 (Mean and variance of the uniform)
For $X \sim \mathrm{Unif}(\alpha, \beta)$, show $\mathbb{E}[X] = \frac{\alpha+\beta}{2}$ and $\mathrm{Var}(X) = \frac{(\beta-\alpha)^2}{12}$.
Solution. The mean is

$$\mathbb{E}[X] = \int_\alpha^\beta \frac{x}{\beta - \alpha} \, dx = \frac{\beta^2 - \alpha^2}{2(\beta - \alpha)} = \frac{\alpha + \beta}{2},$$

the midpoint, as intuition demands. Computing $\mathbb{E}[X^2] = \frac{\beta^3-\alpha^3}{3(\beta-\alpha)} = \frac{\alpha^2+\alpha\beta+\beta^2}{3}$ and subtracting $(\frac{\alpha+\beta}{2})^2$ gives, after simplification, $\mathrm{Var}(X) = \frac{(\beta-\alpha)^2}{12}$.

Example 4.6 (Waiting for a bus)

You reach a stop at 10:00 knowing the bus arrives at a time $X \sim \mathrm{Unif}(0, 30)$ minutes after 10:00. (a) Find the probability you wait more than 10 minutes. (b) Given that the bus has not come by 10:15, find the probability you wait at least 10 more minutes.

Solution. (a) $\mathbb{P}(X > 10) = \int_{10}^{30} \frac{1}{30}\, dx = \frac{2}{3}$. (b) We want $\mathbb{P}(X \geq 25 \,|\, X > 15)$. By definition of conditional probability and since $\{X \geq 25\} \subseteq \{X > 15\}$,

$$\mathbb{P}(X \geq 25 \,|\, X > 15) = \frac{\mathbb{P}(X \geq 25)}{\mathbb{P}(X > 15)} = \frac{1/6}{1/2} = \tfrac{1}{3}.$$

Unlike the exponential below, the uniform is *not* memoryless: waiting already reshapes the remaining wait.

4.3 The Normal Distribution

The *normal* (Gaussian) distribution $\mathcal{N}(\mu, \sigma^2)$ has the bell-shaped density

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}\, e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2},$$

with mean μ and variance σ^2. It is the most important distribution in statistics, partly because of the central limit theorem (Chapter 6). Probabilities are found by standardizing—subtracting μ and dividing by σ to obtain a standard normal $Z \sim \mathcal{N}(0, 1)$—and consulting a table or software.

Example 4.7 (Setting a specification limit)

A lightbulb's output is $\mathcal{N}(2000, 85^2)$ end-foot-candles. Find a lower limit L so that only 5% of bulbs fall below it, i.e. $\mathbb{P}(X \geq L) = 0.95$.

Solution. We need the 5th percentile of the distribution. Since the 5th percentile of a standard normal is -1.645, $L = 2000 - 1.645(85) \approx 1860$. Only about 5% of bulbs output less than 1860.

Example 4.8 (Bolts out of specification)

A machine makes bolts with diameter $\mathcal{N}(1.20, 0.005^2)$ inches; the specification is 1.19 to 1.21 inches. What fraction fail?

Solution. The limits sit exactly two standard deviations either side of the mean, since $(1.21 - 1.20)/0.005 = 2$. The probability of landing outside $\mu \pm 2\sigma$ is about 0.0455, so roughly 4.5% of bolts are out of specification.

Example 4.9 (Chips, and a binomial built on a normal)

Computer-chip lifetimes are $\mathcal{N}(4.4 \times 10^6, (3 \times 10^5)^2)$ hours. (a) A buyer wants at least 90% of chips to last 4.0×10^6 hours; is the supplier acceptable? (b) In a batch of 100 chips, what is the chance at least 4 last less than 3.8×10^6 hours?

Solution. (a) Standardizing, $\mathbb{P}(X \geq 4.0 \times 10^6) \approx 0.909$, so just over 90% qualify—acceptable. (b) First $\mathbb{P}(X < 3.8 \times 10^6) \approx 0.023$. Now let $Y \sim \mathrm{Bin}(100, 0.023)$ count short-lived chips in the batch; then

$$\mathbb{P}(Y \geq 4) = 1 - \sum_{k=0}^{3} \binom{100}{k} (0.023)^k (0.977)^{100-k} \approx 0.199.$$

This example shows the two models cooperating: the normal supplies the per-chip probability, which then drives a binomial count over the batch.

4.4 The Exponential Distribution and Memorylessness

The *exponential* distribution $\mathrm{Exp}(\lambda)$ models waiting times for events that occur at a constant rate λ, with density $f(x) = \lambda e^{-\lambda x}$ for $x \geq 0$ and cdf $F(x) = 1 - e^{-\lambda x}$. Its mean is $1/\lambda$; if a process has mean waiting time μ, then $\lambda = 1/\mu$.

Example 4.10 (The mean of an exponential)

Show that $X \sim \mathrm{Exp}(\lambda)$ has $\mathbb{E}[X] = 1/\lambda$.

Solution. Integration by parts with $u = x$, $dv = \lambda e^{-\lambda x} dx$ gives $\mathbb{E}[X] = \left[-xe^{-\lambda x}\right]_0^\infty + \int_0^\infty e^{-\lambda x}\, dx = 1/\lambda$, as computed in Chapter 1.

Example 4.11 (Finding a median)

Find the median of X with density $f(x) = e^{-x}$, $x \geq 0$ (i.e. $\mathrm{Exp}(1)$).

Solution. Solve $\int_0^m e^{-x}\, dx = \frac{1}{2}$, i.e. $1 - e^{-m} = \frac{1}{2}$, giving $e^{-m} = \frac{1}{2}$ and $m = \ln 2 \approx 0.693$. Note the median is smaller than the mean 1: the exponential is right-skewed.

The exponential's defining feature is *memorylessness*: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$. A used item, conditioned on having survived, is statistically as good as new.

Example 4.12 (Repair times)

The time to repair a machine is $\mathrm{Exp}(\frac{1}{2})$ (mean 2 hours). (a) Find $\mathbb{P}(X > 2)$. (b) Find the conditional probability the repair lasts at least 3 hours given it has already exceeded 2.

Solution. (a) $\mathbb{P}(X > 2) = e^{-\frac{1}{2}\cdot 2} = e^{-1} \approx 0.368$. (b) By memorylessness, $\mathbb{P}(X > 3 \mid X > 2) = \mathbb{P}(X > 1) = e^{-1/2} \approx 0.607$.

Example 4.13 (A used radio)

A radio's lifetime is $\mathrm{Exp}(\frac{1}{8})$ (mean 8 years). You buy it *used*, not knowing its age. What is the probability it lasts another 10 years?

Solution. It seems we lack information—how old is the radio?—but memorylessness makes the age irrelevant. The answer is simply $\mathbb{P}(X \geq 10) = e^{-10/8} = e^{-5/4} \approx 0.287$.

4.5 Heavy Tails: The Pareto Distribution

The *Pareto* distribution $\mathrm{Par}(\alpha)$ has density $f(x) = \alpha/x^{\alpha+1}$ for $x \geq 1$. Its slowly decaying ("heavy") tail can make moments fail to exist—a phenomenon worth seeing at least once.

Example 4.14 (Median of a Pareto)

Find the median of a $\mathrm{Par}(1)$ distribution, whose density is $f(x) = 1/x^2$ for $x \geq 1$.

Solution. Solve $\int_1^m x^{-2}\,dx = \frac{1}{2}$. Since the integral is $1 - \frac{1}{m}$, we need $1 - \frac{1}{m} = \frac{1}{2}$, so $m = 2$.

Example 4.15 (A finite mean but infinite variance)

Let X have density $f(x) = 2/x^3$ for $x \geq 1$ (this is $\mathrm{Par}(2)$). Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution. The mean converges: $\mathbb{E}[X] = \int_1^\infty x \cdot \frac{2}{x^3}\,dx = \int_1^\infty 2x^{-2}\,dx = 2$. But

$$\mathbb{E}[X^2] = \int_1^\infty x^2 \cdot \tfrac{2}{x^3}\,dx = \int_1^\infty \tfrac{2}{x}\,dx = [2\ln x]_1^\infty = \infty,$$

so the variance is *infinite*. Heavy-tailed data can have a well-defined center yet unbounded spread—a warning for any analysis that assumes finite variance.

4.6 Transforming Random Variables

To find the distribution of $Y = g(X)$ for an increasing g, work with cdfs:

$$F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(X \leq g^{-1}(y)) = F_X(g^{-1}(y)).$$

Example 4.16 (Rescaling a uniform)

Let $U \sim \mathrm{Unif}(0,1)$. Find the distribution of $V = rU + s$ for constants $r > 0$, s.

Solution. For $s \leq v \leq r + s$,

$$F_V(v) = \mathbb{P}(rU + s \leq v) = \mathbb{P}\Big(U \leq \tfrac{v-s}{r}\Big) = \tfrac{v-s}{r},$$

which is the cdf of a $\mathrm{Unif}(s,\, r+s)$ distribution. A linear change of scale turns one uniform into another. (For instance $V = 2U + 7$ is $\mathrm{Unif}(7,9)$.)

Example 4.17 (Rescaling an exponential)

Let $X \sim \mathrm{Exp}(\lambda)$. Show that $Y = \lambda X \sim \mathrm{Exp}(1)$.

Solution. Using $F_X(x) = 1 - e^{-\lambda x}$,

$$F_Y(y) = \mathbb{P}(\lambda X \leq y) = \mathbb{P}(X \leq y/\lambda) = 1 - e^{-\lambda(y/\lambda)} = 1 - e^{-y},$$

the cdf of Exp(1). Multiplying an exponential by its own rate produces the standard “rate-one” exponential.

Example 4.18 (Building the Weibull)

Let $X \sim \text{Exp}(1)$ and let $\alpha, \lambda > 0$. Find the cdf of $W = X^{1/\alpha}/\lambda$.

Solution. From $F_X(x) = 1 - e^{-x}$,

$$F_W(w) = \mathbb{P}\Big(\frac{X^{1/\alpha}}{\lambda} \le w\Big) = \mathbb{P}\big(X \le (\lambda w)^\alpha\big) = 1 - e^{-(\lambda w)^\alpha}, \qquad w > 0.$$

This is the *Weibull* distribution with parameters α and λ, widely used in reliability analysis because its shape parameter α lets the failure rate increase or decrease over time.

CHAPTER 5

Joint Distributions, Covariance, and Correlation

So far we have studied one random variable at a time. But many questions concern *two or more* quantities together: hair color and eye color, a chip's defect status across a batch, a wood's density and its hardness. The *joint distribution* encodes how a pair of random variables behave simultaneously, and from it we can read off whether they move together, and how strongly.

5.1 Joint and Marginal Distributions

For discrete X and Y, the *joint pmf* is $p(a,b) = \mathbb{P}(X=a, Y=b)$. From it we recover each variable's own (*marginal*) distribution by summing out the other:

$$p_X(a) = \sum_b p(a,b), \qquad p_Y(b) = \sum_a p(a,b).$$

The marginals live in the row and column totals of the joint table.

Example 5.1 (Completing a joint table)
The joint distribution of $X \in \{0,1,2\}$ and $Y \in \{-1,1\}$ is partly given: the margins are $p_X = (\frac{1}{6}, \frac{2}{3}, \frac{1}{6})$ and $p_Y = (\frac{1}{2}, \frac{1}{2})$, the row $Y=1$ reads $(0, \frac{1}{2}, 0)$. Complete the table and decide whether X and Y are independent.

Solution. Each column must sum to its X-margin, so the row $Y=-1$ is $(\frac{1}{6}, \frac{1}{6}, \frac{1}{6})$:

	0	1	2	p_Y
$Y=-1$	1/6	1/6	1/6	1/2
$Y=1$	0	1/2	0	1/2
p_X	1/6	2/3	1/6	1

They are *dependent*: for instance $\mathbb{P}(X=0, Y=1) = 0 \neq \frac{1}{6} \cdot \frac{1}{2} = \mathbb{P}(X=0)\mathbb{P}(Y=1)$.

Example 5.2 (Reconstructing a joint distribution)
X and Y each take values $1, \ldots, 5$ with margins $p_X = (\frac{1}{14}, \frac{5}{14}, \frac{4}{14}, \frac{2}{14}, \frac{2}{14})$ and $p_Y = (\frac{5}{14}, \frac{4}{14}, \frac{2}{14}, \frac{2}{14}, \frac{1}{14})$. Each joint probability is either 0 or $\frac{1}{14}$. Find the joint distribution.

Solution. Each cell is $\frac{1}{14}$ or 0, and the number of $\frac{1}{14}$ cells in a row (column) must equal 14 times that row's (column's) margin. Working row by row—five entries in the row summing to $\frac{5}{14}$, four in the next, and so on, while respecting the column totals—forces a unique "staircase" pattern in which row i has $14p_Y(i)$ filled cells packed toward the columns with the largest remaining demand. The filled cells are exactly those (a, b) for which the cumulative column and row counts still admit a unit, producing the table whose row sums are $5, 4, 2, 2, 1$ (times $\frac{1}{14}$) and column sums $1, 5, 4, 2, 2$.

Example 5.3 (Hair color and eye color)

The hair and eye color of 5383 people are cross-classified. Dividing the counts by 5383 turns the table into a joint distribution for hair color X and eye color Y. Are X and Y independent?

Solution. The joint and marginal probabilities are the cell counts and row/column totals divided by 5383. Independence fails dramatically: the observed $\mathbb{P}(\text{fair hair, light eyes}) = 1168/5383 \approx 0.22$, but the product of margins is $(1741/5383)(2298/5383) \approx 0.14$. Fair hair and light eyes co-occur far more often than independence would predict—the variables are strongly associated.

5.2 Expectation of a Function; Covariance and Correlation

For a function $g(X, Y)$, $\mathbb{E}[g(X, Y)] = \sum_{a,b} g(a, b)\, p(a, b)$. The most important case is the *covariance*, which measures how X and Y vary together:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Its scale-free version is the *correlation coefficient*

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} \in [-1, 1].$$

Independent variables are uncorrelated ($\rho = 0$), but the converse can fail.

Example 5.4 (Covariance from a joint table)

For the joint distribution of U, V with $\mathbb{E}[U] = 1$, $\mathbb{E}[V] = \frac{1}{2}$, and $\mathbb{E}[UV] = \frac{1}{2}$, compute $\text{Cov}(U, V)$ and $\rho(U, V)$.

Solution. $\text{Cov}(U, V) = \mathbb{E}[UV] - \mathbb{E}[U]\mathbb{E}[V] = \frac{1}{2} - (1)(\frac{1}{2}) = 0$. Since both variances are positive, $\rho(U, V) = 0$ as well. Here zero covariance happens to coincide with the variables being uncorrelated by construction.

Example 5.5 (Uncorrelated but dependent, and variances of sums)

Consider $X \in \{0, 1, 2\}$, $Y \in \{0, 1\}$ with joint distribution

	0	1	2	p_Y
$Y = 0$	1/6	1/6	1/6	1/2
$Y = 1$	1/4	0	1/4	1/2
p_X	5/12	1/6	5/12	1

Show X and Y are dependent yet uncorrelated, and find $\text{Var}(X + Y)$ and $\text{Var}(X - Y)$.

Solution. One computes $\mathbb{E}[XY] = \frac{1}{2}$, $\mathbb{E}[X] = 1$, $\mathbb{E}[Y] = \frac{1}{2}$, so $\text{Cov}(X, Y) = \frac{1}{2} - (1)(\frac{1}{2}) = 0$: uncorrelated. Yet they are dependent, since $\mathbb{P}(X = 1, Y = 1) = 0 \neq \mathbb{P}(X = 1)\mathbb{P}(Y = 1)$. Thus *zero correlation does not imply independence*—correlation only detects *linear* association. With $\mathrm{Var}(X) = \frac{5}{6}$ and $\mathrm{Var}(Y) = \frac{1}{4}$, and using

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\mathrm{Cov}(X, Y),$$

both $\mathrm{Var}(X+Y)$ and $\mathrm{Var}(X-Y)$ equal $\frac{5}{6} + \frac{1}{4} = \frac{13}{12}$, because the covariance term vanishes.

Example 5.6 (Tuning the correlation)

Let $X, Y \in \{0, 1\}$ have joint distribution depending on a parameter ϵ with $-\frac{1}{4} \le \epsilon \le \frac{1}{4}$:

	0	1	p_X
0	$1/4-\epsilon$	$1/4+\epsilon$	$1/2$
1	$1/4+\epsilon$	$1/4-\epsilon$	$1/2$
p_Y	$1/2$	$1/2$	1

Express $\rho(X, Y)$ in terms of ϵ and find the values giving $\rho = 1, 0, -1$.

Solution. The margins are uniform on $\{0, 1\}$, so $\mathbb{E}[X] = \mathbb{E}[Y] = \frac{1}{2}$ and $\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{1}{4}$ regardless of ϵ. Computing $\mathbb{E}[XY] = \frac{1}{4} - \epsilon$ gives $\mathrm{Cov}(X, Y) = (\frac{1}{4} - \epsilon) - \frac{1}{4} = -\epsilon$, hence

$$\rho(X, Y) = \frac{-\epsilon}{\sqrt{(1/4)(1/4)}} = -4\epsilon.$$

So $\rho = 1$ when $\epsilon = -\frac{1}{4}$ (the off-diagonal probabilities vanish and $X = Y$ deterministically), $\rho = 0$ when $\epsilon = 0$ (independence), and $\rho = -1$ when $\epsilon = \frac{1}{4}$ ($X = 1 - Y$). The single parameter ϵ sweeps the correlation across its entire range.

5.3 Continuous Joint Distributions

For continuous X, Y the joint density $f(x, y)$ satisfies $\mathbb{P}((X, Y) \in A) = \iint_A f\,dx\,dy$, the marginals are $f_X(x) = \int f(x, y)\,dy$ and $f_Y(y) = \int f(x, y)\,dx$, and independence is $f(x, y) = f_X(x)f_Y(y)$ (equivalently $F(x, y) = F_X(x)F_Y(y)$).

Example 5.7 (From a joint cdf)

Suppose $F(x, y) = 1 - \frac{1}{x} - \frac{1}{y^2} + \frac{1}{xy^2}$ for $x \ge 1$, $y \ge 1$ (and 0 otherwise). Find the joint density, the marginal densities, and decide whether X and Y are independent.

Solution. The joint density is the mixed partial derivative,

$$f(x, y) = \frac{\partial^2 F}{\partial x\,\partial y} = \frac{2}{x^2 y^3}, \qquad x \ge 1,\ y \ge 1.$$

Integrating out one variable at a time, $f_X(x) = \int_1^\infty \frac{2}{x^2y^3}\,dy = \frac{1}{x^2}$ and $f_Y(y) = \int_1^\infty \frac{2}{x^2y^3}\,dx = \frac{2}{y^3}$. Since $f_X(x)f_Y(y) = \frac{2}{x^2y^3} = f(x, y)$, the variables are *independent*. (Equivalently the cdf factors as $F(x, y) = (1 - \frac{1}{x})(1 - \frac{1}{y^2})$.)

Example 5.8 (A product-form density)

Let $f(x, y) = 12x(1-x)y$ for $0 \le x \le 1$, $0 \le y \le 1$. Find the marginal densities, the joint cdf, and $\mathrm{Cov}(X, Y)$.

Solution. Integrating out y, then x,

$$f_X(x) = \int_0^1 12x(1-x)y\,dy = 6x(1-x), \qquad f_Y(y) = \int_0^1 12x(1-x)y\,dx = 2y.$$

Because $f(x, y) = f_X(x)f_Y(y)$, the variables are independent, and the joint cdf factors:

$$F(x, y) = \left(3x^2 - 2x^3\right)y^2, \qquad 0 \le x, y \le 1.$$

Independence forces $\mathrm{Cov}(X, Y) = 0$. We can confirm directly: $\mathbb{E}[X] = \frac{1}{2}$, $\mathbb{E}[Y] = \frac{2}{3}$, and $\mathbb{E}[XY] = \frac{1}{3} = \mathbb{E}[X]\mathbb{E}[Y]$, so indeed $\mathrm{Cov}(X, Y) = 0$.

5.4 Linearity of Expectation in Action

Expectation is linear even when variables are dependent, and this single fact solves problems that look forbidding.

Example 5.9 (Group testing of blood samples)

The blood of 1000 people must be screened for a disease that each carries independently with probability $p = 0.001$. Rather than 1000 individual tests, pool the samples into 25 groups of 40: test each pooled sample, and if it is positive, retest all 40 individuals (so that group costs 41 tests). Let X_i be the number of tests for group i. Find the expected total number of tests.

Solution. For a group, $X_i = 1$ if the pool is clean and $X_i = 41$ otherwise. The pool is clean with probability $(1 - 0.001)^{40} \approx 0.96$, so $\mathbb{P}(X_i = 1) \approx 0.96$ and $\mathbb{P}(X_i = 41) \approx 0.04$. Then

$$\mathbb{E}[X_i] = 1(0.96) + 41(0.04) = 2.6,$$

and by linearity the expected total over all 25 groups is $\mathbb{E}\left[\sum_{i=1}^{25} X_i\right] = 25(2.6) = 65$. Pooling replaces 1000 guaranteed tests with about 65 expected tests—a striking economy that exploits the rarity of the disease.

CHAPTER 6

Inequalities and Limit Theorems

This chapter collects three of the deepest and most useful results about averages of random variables: Jensen's inequality, which compares the average of a function to the function of the average; Chebyshev's inequality, which bounds how far a random variable can stray from its mean; and the law of large numbers and central limit theorem, which together explain why sample averages stabilize and why the normal distribution is ubiquitous.

6.1 Jensen's Inequality

A function g is *convex* if its graph lies below its chords (equivalently $g'' \geq 0$), and *concave* if the reverse holds. *Jensen's inequality* says that for a convex g,

$$g(\mathbb{E}[X]) \leq \mathbb{E}[g(X)],$$

with the inequality reversed for concave g. The averaging and the bending interact in a fixed direction.

Example 6.1

For a positive random variable X, which is larger, $\mathbb{E}[\ln X]$ or $\ln(\mathbb{E}[X])$?

Solution. The logarithm is concave, so Jensen's inequality (reversed) gives $\ln(\mathbb{E}[X]) \geq \mathbb{E}[\ln X]$. Taking the logarithm of an average overstates the average of the logarithms.

Example 6.2

For a positive random variable X, which is larger, $\mathbb{E}[e^X]$ or $e^{\mathbb{E}[X]}$?

Solution. The exponential is convex, so Jensen's inequality gives $\mathbb{E}[e^X] \geq e^{\mathbb{E}[X]}$. The two examples illustrate how the direction of the inequality follows the curvature of the function.

We will use Jensen's inequality again in Chapter 9 to show that certain natural estimators are systematically biased.

6.2 Chebyshev's Inequality

Chebyshev's inequality bounds the probability that a random variable lands far from its mean, using only the variance:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}, \qquad a > 0.$$

It requires no knowledge of the distribution's shape, which makes it both very general and rather conservative.

Example 6.3 (Bounding a test score)

A final-exam score X has mean 75 and variance 25. What can be said about $\mathbb{P}(55 \le X \le 95)$?

Solution. The interval is 75 ± 20, so we want $\mathbb{P}(|X - 75| \le 20)$. Chebyshev bounds the complementary event: $\mathbb{P}(|X - 75| \ge 20) \le \frac{25}{20^2} = \frac{1}{16}$. Therefore $\mathbb{P}(55 \le X \le 95) \ge 1 - \frac{1}{16} = \frac{15}{16}$. At least 93.75% of scores lie within the stated range, whatever the score distribution looks like.

Example 6.4 (How large a poll?)

To predict the proportion p of voters favoring a candidate, we interview n people and use $\bar{X}_n$, the sample fraction. How large must n be so that $\mathbb{P}(|\bar{X}_n - p| < 0.2) \ge 0.9$?

Solution. Each respondent is $\mathrm{Ber}(p)$, so $\mathbb{E}[\bar{X}_n] = p$ and $\mathrm{Var}(\bar{X}_n) = p(1-p)/n$. By Chebyshev,

$$\mathbb{P}(|\bar{X}_n - p| \ge 0.2) \le \frac{p(1-p)/n}{0.2^2} = \frac{25\,p(1-p)}{n} \le \frac{25}{4n},$$

using $p(1-p) \le \frac{1}{4}$. Requiring this to be at most 0.1 gives $n \ge 62.5$, so $n = 63$ suffices for *every* value of p. (The same calculation with a tolerance of 0.1 instead of 0.2 demands $n \ge 250$, and a confidence of 0.95 instead of 0.9 demands $n \ge 125$.) Chebyshev's distribution-free guarantee is convenient but, as the next section shows, far from tight.

6.3 The Law of Large Numbers

The *law of large numbers* (LLN) makes precise the intuition that averages settle down: if $X_1, X_2, \dots$ are independent with common mean μ and finite variance, then for any $\epsilon > 0$,

$$\mathbb{P}(|\bar{X}_n - \mu| > \epsilon) \to 0 \quad \text{as } n \to \infty.$$

The sample average converges to the true mean.

Example 6.5 (The casino's edge)

In the game Red-or-Black, the casino wins a \$1 bet with probability $\frac{19}{37}$ and loses \$1 with probability $\frac{18}{37}$. The game is played about 1000 times a night, 365 nights a year. Estimate the casino's annual income from the game.

Solution. Let $X_i = \pm 1$ be the casino's gain on play i, so $\mathbb{E}[X_i] = 1 \cdot \frac{19}{37} + (-1) \cdot \frac{18}{37} = \frac{1}{37}$. With $n = 365000$ plays per year, the LLN says the average gain per play is, with overwhelming probability, near $\frac{1}{37}$. By linearity the expected annual total is

$$\mathbb{E}\Big[\sum_{i=1}^{n} X_i\Big] = n \cdot \tfrac{1}{37} = 365000 \cdot \tfrac{1}{37} \approx \$9865.$$

The casino's edge is tiny per play, but the law of large numbers turns it into a dependable income.

6.4 The Central Limit Theorem

The *central limit theorem* (CLT) says that the *sum* (or average) of many independent, identically distributed random variables is approximately normal, regardless of the original distribution: if each X_i has mean μ and variance σ^2, then for large n,

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \;\approx\; \mathcal{N}(0,1).$$

This single fact underlies most of classical statistics.

Example 6.6 (Defective chips in a batch)
Each chip is defective with probability $\frac{1}{5}$, independently. In a batch of 100, approximate the probability that fewer than 15 are defective.

Solution. Let $X_i \sim \mathrm{Ber}(\frac{1}{5})$, so $\mathbb{E}[X_i] = \frac{1}{5}$ and $\mathrm{Var}(X_i) = \frac{4}{25}$. The total $S = \sum X_i$ has mean 20 and variance 16, so $\mathrm{sd}(S) = 4$. By the CLT,

$$\mathbb{P}(S < 15) \approx \mathbb{P}\Big(Z < \tfrac{15-20}{4}\Big) = \mathbb{P}(Z < -1.25) = \mathbb{P}(Z > 1.25) \approx 0.106.$$

Example 6.7 (A sum from an unfamiliar density)
Let $X_1, \ldots, X_{625}$ be independent with density $f(x) = 3(1-x)^2$ on $[0,1]$. Approximate $\mathbb{P}(X_1 + \cdots + X_{625} < 170)$.

Solution. First find the moments. Integrating, $\mathbb{E}[X_i] = \int_0^1 x \cdot 3(1-x)^2\,dx = \frac{1}{4}$ and $\mathbb{E}[X_i^2] = \int_0^1 x^2 \cdot 3(1-x)^2\,dx = \frac{1}{10}$, so $\mathrm{Var}(X_i) = \frac{1}{10} - \frac{1}{16} = \frac{3}{80}$. The sum has mean $625 \cdot \frac{1}{4} = 156.25$ and standard deviation $\sqrt{3/80} \cdot 25 \approx 4.84$. By the CLT,

$$\mathbb{P}(S < 170) \approx \mathbb{P}\Big(Z < \tfrac{170-156.25}{4.84}\Big) \approx \mathbb{P}(Z < 2.84) \approx 0.998.$$

Example 6.8 (Chain links, and the danger of a small bias)
Links have length with mean 5 cm and variance 0.04. A chain of 1002 links is sold as "at least 50 m." (a) What fraction of chains fall short? (b) If the true mean is actually 4.99 cm, what fraction fall short?

Solution. (a) A chain is short when the link total is below 5000 cm. With mean $1002 \cdot 5 = 5010$ and standard deviation $\sqrt{1002 \cdot 0.04} \approx 6.33$,

$$\mathbb{P}(S < 5000) \approx \mathbb{P}\Big(Z < \tfrac{5000-5010}{6.33}\Big) \approx 0.057,$$

about 6%. (b) With mean 4.99 the total mean is 5000.0 (nearly exactly 5000), so $\mathbb{P}(S < 5000) \approx \frac{1}{2}$: about half the chains fall short. A seemingly negligible change in the mean—one part in five hundred—roughly *octuples* the give-away rate. Small biases, compounded over many links, have large consequences.

Example 6.9 (A sum of squared normals)
Let $X_1, X_2, \ldots$ be independent $\mathcal{N}(0,1)$ and set $Y_n = X_1^2 + \cdots + X_n^2$. Given that $\mathbb{E}[X_i^2] = 1$ and $\mathbb{E}[X_i^4] = 3$, approximate $\mathbb{P}(Y_{50} > 60)$.

Solution. Each squared term has mean 1 and variance $\mathbb{E}[X_i^4] - (\mathbb{E}[X_i^2])^2 = 3 - 1 = 2$. Thus Y_{50} has mean 50 and variance 100, so standard deviation 10. By the CLT,

$$\mathbb{P}(Y_{50} > 60) \approx \mathbb{P}\Big(Z > \tfrac{60-50}{10}\Big) = \mathbb{P}(Z > 1) \approx 0.159.$$

The variable Y_n is a *chi-square* random variable with n degrees of freedom, a distribution we will meet again in Chapters 8 and 12.

6.4.1 Chebyshev versus the central limit theorem

Example 6.10 (Revisiting the poll)
Using the CLT instead of Chebyshev, how large must the poll be so that $\mathbb{P}(|\bar{X}_n - p| < 0.2) \geq 0.9$?

Solution. Standardizing, the requirement becomes $\mathbb{P}\big(|Z| < \sqrt{n}\,\frac{0.2}{\sqrt{p(1-p)}}\big) \geq 0.9$, i.e. $\sqrt{n}\,\frac{0.2}{\sqrt{p(1-p)}} \geq 1.645$. Using $p(1-p) \leq \frac{1}{4}$ in the worst case yields $n \geq 17$. Compare this with the $n \geq 63$ demanded by Chebyshev: the CLT, by using the *shape* of the distribution and not just its variance, gives a far sharper—and in practice more realistic—sample size.

CHAPTER 7

Exploratory Data Analysis

Until now we have built probability models for random phenomena. When we meet a new phenomenon, though, we usually start from the other end: we collect a set of observations—a *dataset*—and try to learn what kind of randomness produced it. *Exploratory data analysis* (EDA) is the art of summarizing a dataset, graphically and numerically, so that its essential features stand out. These summaries are *descriptive*: they let the data speak for themselves, without yet inferring anything about an underlying distribution.

Throughout this chapter we use as a running example the durations (in seconds) of 272 eruptions of the Old Faithful geyser. Simply listing the numbers, or even sorting them, reveals little; we need better summaries.

7.1 Graphical Summaries

7.1.1 The histogram

A *histogram* divides the range of the data into intervals, or *bins* $B_1, \ldots, B_m$, and draws a bar over each. We scale the histogram so the total area is one, which makes it a crude estimate of an underlying density: the height over bin B_i is

$$\text{height} = \frac{\text{number of observations in } B_i}{n\,|B_i|},$$

where $|B_i|$ is the bin width. The area of each bar then equals the proportion of data in that bin.

The choice of bin width is a genuine tradeoff. Too narrow, and the histogram dissolves into a jagged row of isolated spikes; too wide, and real structure is smoothed away. Figure 7.1 shows the geyser data at three bin widths. The middle choice reveals the dataset's most important feature—*two* clusters of eruption durations, near 120 and 270 seconds—that the widest bins erase entirely.

7.1.2 Kernel density estimates

A histogram is simple but jagged, and sensitive to where the bin edges fall. A *kernel density estimate* (KDE) produces a smooth curve instead. The idea is to "place a small pile of sand" over each observation and add the piles up; where data accumulate, the sand piles high. Formally one chooses a *kernel* K—a symmetric density giving each pile its shape—and a *bandwidth* h controlling each pile's width.

A kernel K typically satisfies $K(u) \geq 0$, integrates to one, is symmetric about zero, and vanishes outside $[-1, 1]$. Common choices include the Epanechnikov kernel $K(u) = \frac{3}{4}(1 - u^2)$ and the triweight kernel $K(u) = \frac{35}{32}(1 - u^2)^3$; the normal kernel is also popular, though it violates the last condition. Figure 7.3 compares several.

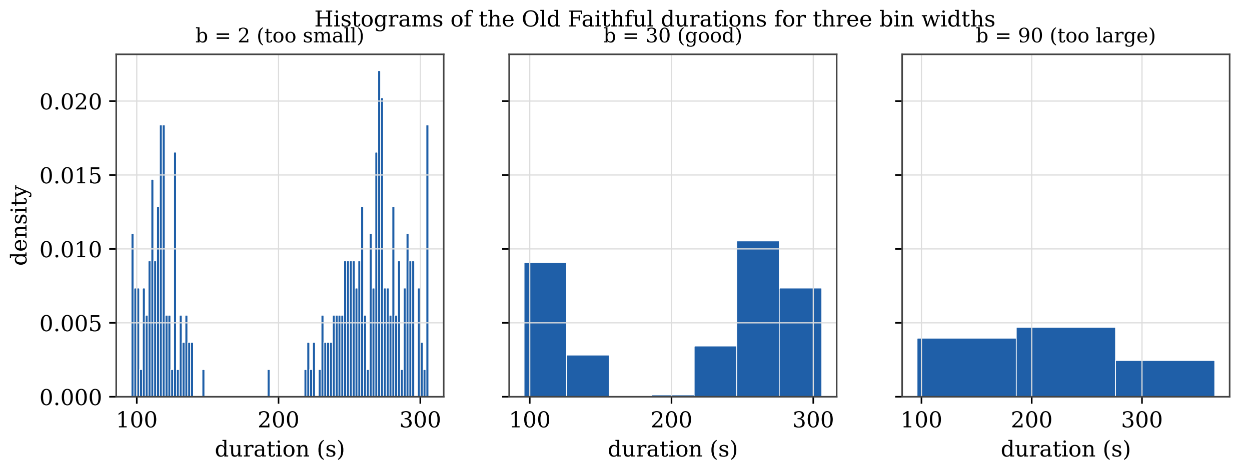


Figure 7.1. The same geyser data at three bin widths. Too small produces noise; too large hides the two modes. In practice one chooses the bin width by trial and error until the picture looks reasonable.

The bandwidth plays exactly the role the bin width played for the histogram: too small gives a spiky curve, too large oversmooths and can wash out the modes (Figure 7.4). The choice of *kernel*, by contrast, has only a minor effect on the result.

7.1.3 The empirical distribution function

A third graphical summary plots the data cumulatively. The *empirical distribution function* (or empirical cdf) is

$$F_n(x) = \frac{\text{number of observations} \leq x}{n},$$

the proportion of the dataset at or below x. Its graph is a staircase: flat below the minimum, equal to one above the maximum, and rising by a jump of $1/n$ at each observation.

> **Example 7.1 (A small empirical cdf)**
> For the dataset $\{4, 3, 9, 1, 7\}$, the empirical cdf is 0 for $x < 1$, jumps to $\frac{1}{5}$ at $x = 1$, to $\frac{2}{5}$ at $x = 3$, to $\frac{3}{5}$ at $x = 4$, and so on up to 1 for $x \geq 9$ (Figure 7.5). The closed and open circles emphasize that the function takes its *higher* value at each jump point.

For the geyser data (Figure 7.6), the empirical cdf rises *steeply* where observations crowd together—near 120 and 270 seconds—and flattens where they thin out. Steepness of F_n is the cumulative signature of a mode.

7.1.4 The scatterplot

For a *bivariate* dataset of pairs $(x_1, y_1), \ldots, (x_n, y_n)$, the *scatterplot* simply plots the points. It is the natural first step when we suspect that y depends on x. We will return to scatterplots in Chapter 11, where we fit lines to them; see, for instance, the timber and mortality data in that chapter.

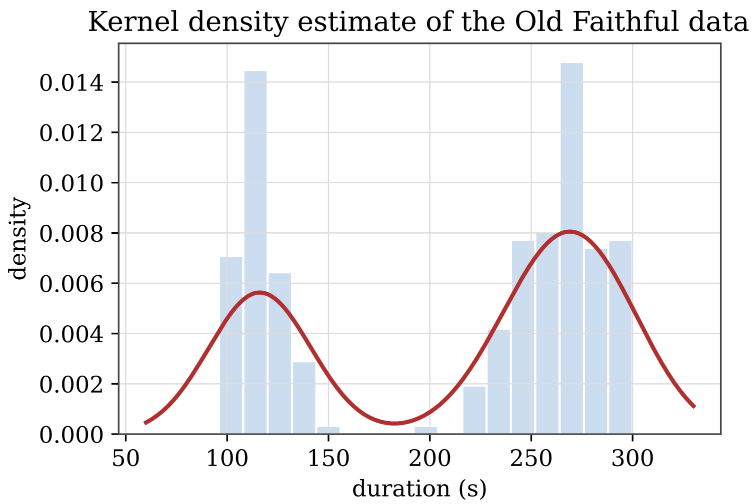


Figure 7.2. A kernel density estimate of the geyser data overlaid on a histogram. The two modes are now unmistakable and the curve is smooth.

Example 7.2 (Correlation of geyser eruptions and waiting times)
For the geyser, each eruption's duration can be paired with the waiting time until the next eruption. The empirical correlation coefficient of these two quantities is about 0.90—a strong positive association: long eruptions tend to be followed by long waits.

Example 7.3 (Height and weight, and a prediction)
For a classic dataset of women's heights and weights, fitting a line gives height $\approx 0.287\,(\text{weight})+25.72$ inches, with an empirical correlation of 0.995—an exceptionally strong linear relationship. Using the line, a woman weighing 105 pounds is predicted to be about $0.287(105) + 25.72 \approx 56$ inches tall.

Example 7.4 (Linearizing exponential growth)
The U.S. population appears to grow roughly exponentially, $P(t) = ae^{kt}$. Taking logarithms turns this into a *linear* relationship, $\ln P(t) = \ln a + kt$, so regressing $\ln P$ on t recovers the parameters. With t measured in years from 1790, this yields $\ln a \approx 1.688$ (so $a \approx 5.41$ million) and $k \approx 0.0220$, i.e. $P(t) \approx 5.41\, e^{0.0220t}$. Transforming to linearity before fitting is a recurring EDA tactic.

7.2 Numerical Summaries

Graphs reveal shape; numbers pin down location and spread.

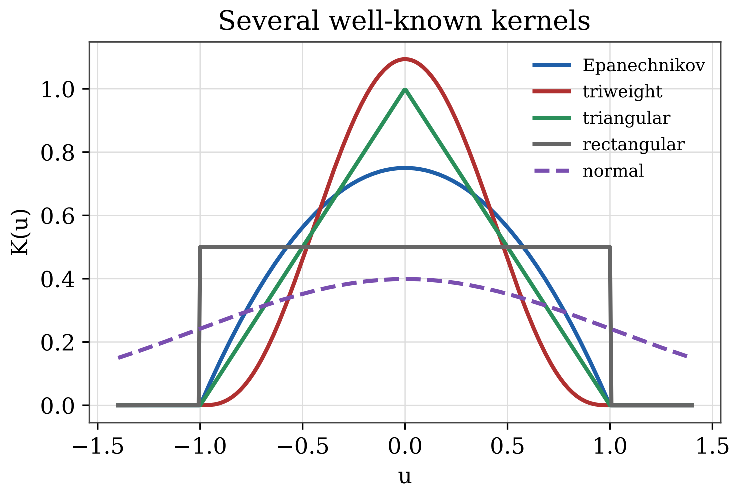


Figure 7.3. Several standard kernels. The choice of kernel matters far less than the choice of bandwidth.

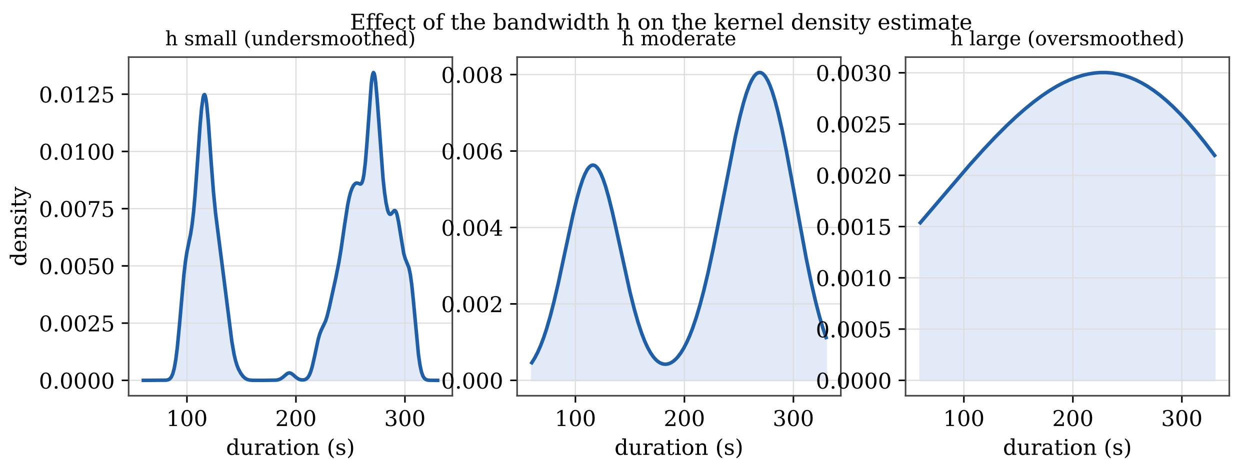


Figure 7.4. The bandwidth h controls smoothness. Two well-known kernels with the same bandwidth produce nearly identical curves; the bandwidth is what matters.

7.2.1 The center of a dataset

The *sample mean* is the ordinary average $\bar{x}_n = \frac{1}{n}\sum_{i=1}^{n} x_i$, the natural analogue of expectation. The *sample median* is the middle value of the sorted data (the average of the two middle values when n is even). The mean is sensitive to *outliers*; the median is robust.

Example 7.5 (Outliers and a recording error)

Eleven hourly temperatures recorded one New Year's at Wick, Scotland, were

$$43, 43, 41, 41, 41, 42, 43, 58, 58, 41, 41.$$

The two 58's sit far from the rest and inflate the mean to 44.7; dropping them lowers it to 41.8. The median barely moves (42 versus 41). In fact the two anomalies arose from a switch of recording units; the corrected values are near 41, giving mean 41.5 and median 42. The

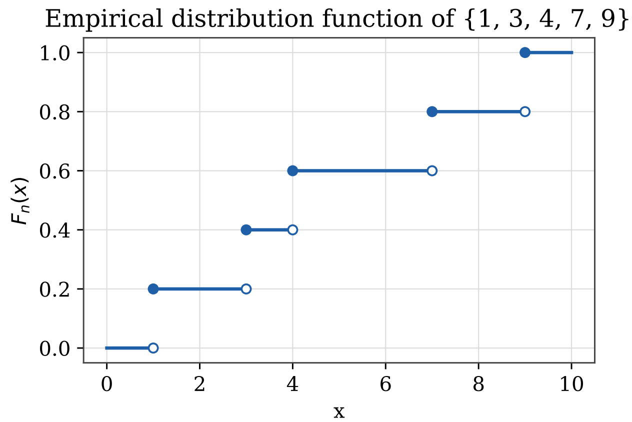


Figure 7.5. The empirical distribution function of a five-point dataset.

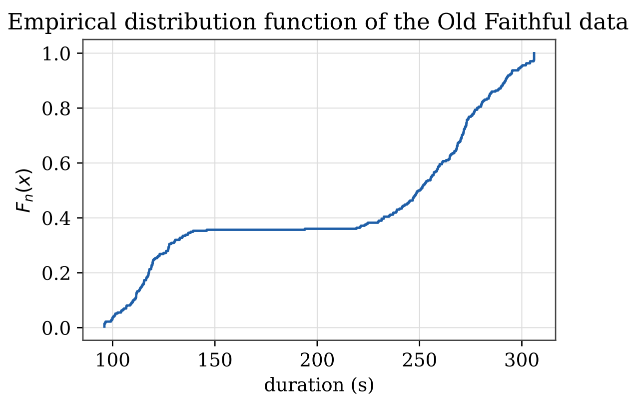


Figure 7.6. Empirical distribution function of the geyser data; the two steep stretches correspond to the two clusters of durations.

> median's robustness is exactly its insensitivity to such contamination. (This is an argument for being *aware* of outliers, not for blindly deleting them.)

7.2.2 The spread of a dataset

The *sample variance* and *sample standard deviation* are

$$s_n^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x}_n)^2, \qquad s_n = \sqrt{s_n^2}.$$

The divisor $n-1$ rather than n will be justified in Chapter 9, where we show it makes s_n^2 unbiased. Like the mean, the standard deviation is sensitive to outliers: for the uncorrected Wick data $s_n = 6.62$, but only 0.97 with the two anomalies removed.

A robust alternative is the *median of absolute deviations*,

$$\mathrm{MAD}(x_1, \ldots, x_n) = \mathrm{Med}\big(|x_1 - \mathrm{Med}_n|, \ldots, |x_n - \mathrm{Med}_n|\big),$$

which, like the median, is hardly affected by a few extreme values.

7.2.3 Quantiles, quartiles, and the IQR

The *pth empirical quantile* $q_n(p)$ is, loosely, the value below which a proportion p of the data lie. Writing the sorted data (the *order statistics*) as $x_{(1)} \leq \cdots \leq x_{(n)}$, we take the ith order statistic to be the $\frac{i}{n+1}$ quantile and interpolate linearly between them.

Example 7.6 (Interpolating a percentile)
For the Wick data, the order statistics put 42 at the $\frac{6}{12} = 0.5$ quantile and 43 at the $\frac{7}{12}$ quantile. The 55th percentile lies between these; linear interpolation between $(0.5, 42)$ and $(0.583, 43)$ gives $q_n(0.55) \approx 42.6$.

The lower and upper *quartiles* are $q_n(0.25)$ and $q_n(0.75)$, and the *interquartile range* is

$$\mathrm{IQR} = q_n(0.75) - q_n(0.25),$$

the span of the middle half of the data—another robust measure of spread. Together with the minimum, median, and maximum, the quartiles form the *five-number summary*.

Example 7.7 (A five-number summary)
For 1000 recorded earthquake magnitudes, the five-number summary is

$$\min = 4.00, \quad Q_1 = 4.30, \quad \text{median} = 4.60, \quad Q_3 = 4.90, \quad \max = 6.40,$$

with sample mean 4.62. The closeness of mean and median, and the symmetry of the quartiles about the median, suggest a roughly symmetric bulk with a modest right tail.

7.2.4 The boxplot

The *boxplot* renders the five-number summary graphically (Figure 7.7). A box spans the lower to upper quartile—its height is the IQR—with a line at the median. *Whiskers* extend to the most extreme observations within $1.5 \times \mathrm{IQR}$ of the box; anything beyond is plotted individually as a suspected outlier.

The boxplot is superb for comparing spreads and spotting skew and outliers, but it has a blind spot: it cannot show that the geyser data are *bimodal*. The position of the median within the box hints at asymmetry, but only the histogram and KDE reveal the two peaks. No single summary tells the whole story—which is why exploratory analysis uses several in concert.

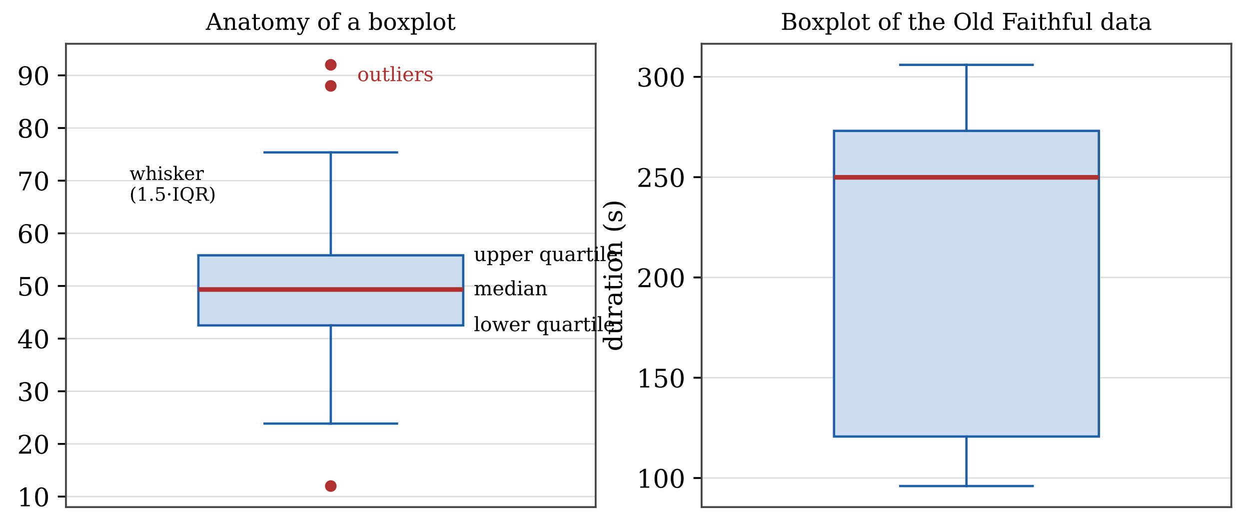


Figure 7.7. Left: the anatomy of a boxplot. Right: a boxplot of the geyser data. Boxplots are excellent for spread and skew but, as here, cannot reveal that a dataset is bimodal—the off-center median is the only hint of structure.

CHAPTER 8

An Overview of Statistical Inference

Exploratory data analysis describes a dataset. *Statistical inference* goes further: it uses the data to draw conclusions about the unknown distribution that generated them. Given a sample $X_1, \dots, X_n \sim F$, we want to infer F, or some feature of it such as its mean or variance. This chapter surveys the three central inference problems—*point estimation*, *confidence intervals*, and *hypothesis testing*—and introduces the t distribution that the later chapters depend on.

8.1 Statistical Models

A *statistical model* is a set $\mathcal{F}$ of candidate distributions. A *parametric* model is one indexed by finitely many parameters, written $\mathcal{F} = \{f(x;\theta) : \theta \in \Theta\}$, where θ ranges over a parameter space Θ. For example, the normal model

$$\mathcal{F} = \left\{ f(x;\mu,\sigma) = \tfrac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}((x-\mu)/\sigma)^2} \;:\; \mu \in \mathbb{R},\ \sigma > 0 \right\}$$

has two parameters. If only one parameter interests us—say μ—the others are called *nuisance* parameters. A *nonparametric* model assumes no such finite-dimensional form; estimating an unknown cdf by the empirical cdf of Chapter 7 is a nonparametric procedure.

8.2 Point Estimation

A *point estimator* of a parameter θ is a function of the data, $\hat{\theta}_n = g(X_1, \dots, X_n)$, intended as a single best guess. Because it depends on the random sample, $\hat{\theta}_n$ is itself a random variable, with a distribution called its *sampling distribution*. Several quantities measure the quality of an estimator:

- the *bias*, $\mathrm{bias}(\hat{\theta}_n) = \mathbb{E}[\hat{\theta}_n] - \theta$; the estimator is *unbiased* if $\mathbb{E}[\hat{\theta}_n] = \theta$;
- the *standard error* $\mathrm{se} = \sqrt{\mathrm{Var}(\hat{\theta}_n)}$, the standard deviation of the sampling distribution (its estimate is $\widehat{\mathrm{se}}$);
- the *mean squared error* $\mathrm{MSE} = \mathbb{E}[(\hat{\theta}_n - \theta)^2]$.

These are linked by a fundamental identity.

Theorem 8.1. *The mean squared error decomposes as* $\mathrm{MSE} = \left(\mathrm{bias}(\hat{\theta}_n)\right)^2 + \mathrm{Var}(\hat{\theta}_n)$.

An estimator is *asymptotically normal* if $(\hat{\theta}_n - \theta)/\mathrm{se}$ converges to $\mathcal{N}(0,1)$ as $n \to \infty$; many estimators are, thanks to the central limit theorem.

Example 8.1 (Estimating a Bernoulli probability)
Let $X_1, \ldots, X_n \sim \text{Ber}(p)$ and let $\hat{p}_n = \bar{X}_n = \frac{1}{n}\sum X_i$. Then $\mathbb{E}[\hat{p}_n] = \frac{1}{n}\sum \mathbb{E}[X_i] = p$, so $\hat{p}_n$ is unbiased. Its standard error is $\text{se} = \sqrt{\text{Var}(\hat{p}_n)} = \sqrt{p(1-p)/n}$, estimated by $\widehat{\text{se}} = \sqrt{\hat{p}_n(1-\hat{p}_n)/n}$.

8.3 Confidence Intervals

A $1-\alpha$ *confidence interval* for θ is a random interval $C_n = (a, b)$, computed from the data, with

$$\mathbb{P}(\theta \in C_n) \geq 1 - \alpha \quad \text{for all } \theta.$$

We call $1-\alpha$ the *coverage*.

Remark 8.2 (How to read a confidence interval)**.** A confidence interval is *not* a probability statement about θ: θ is a fixed (if unknown) number, not random. The randomness is in the interval. The honest interpretation is a long-run one: if you construct a 95% interval every day—for entirely unrelated parameters—about 95% of those intervals will trap their target. Each individual interval either contains θ or it does not (Figure 8.1). When a newspaper reports a poll "accurate to within 4 points 95% of the time," it is describing exactly this kind of interval.

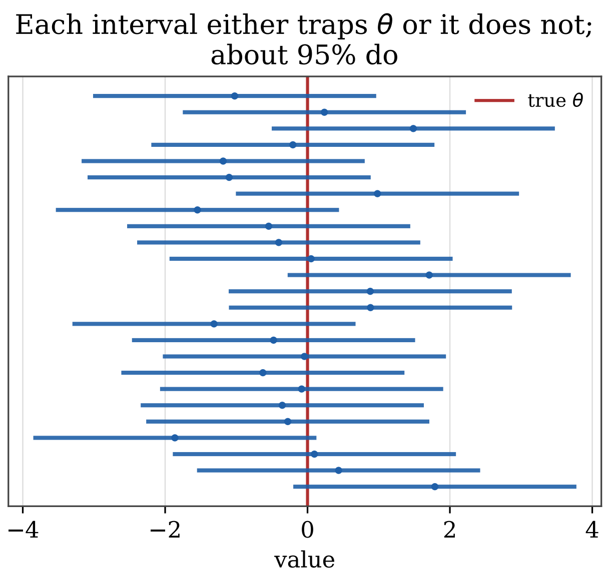


Figure 8.1. Twenty-five 95% confidence intervals for the same θ. About one in twenty misses (shown in red). No single interval "contains θ with probability 0.95"—it either does or does not.

8.3.1 Normal-based intervals

When $\hat{\theta}_n \approx \mathcal{N}(\theta, \widehat{\text{se}}^2)$, we obtain an approximate interval by going $z_{\alpha/2}$ standard errors either side of the estimate:

$$C_n = (\hat{\theta}_n - z_{\alpha/2}\,\widehat{\text{se}},\ \hat{\theta}_n + z_{\alpha/2}\,\widehat{\text{se}}),$$

where $z_{\alpha/2}$ satisfies $\mathbb{P}(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$ for $Z \sim \mathcal{N}(0,1)$ (Figure 8.2). For 95% coverage, $z_{\alpha/2} = 1.96 \approx 2$, giving the rule of thumb $\hat{\theta}_n \pm 2\,\widehat{\mathrm{se}}$.

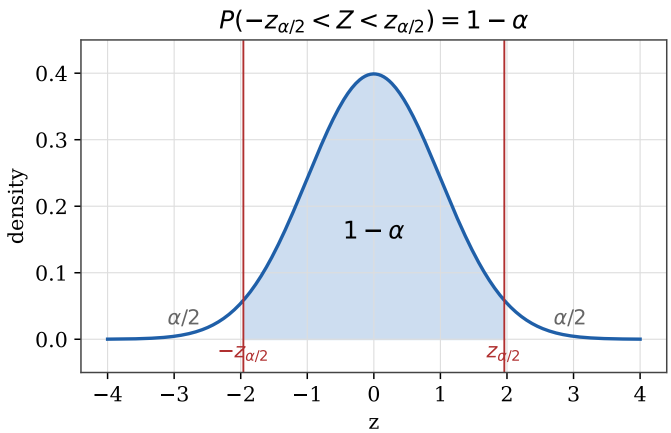


Figure 8.2. The critical value $z_{\alpha/2}$ cuts off probability $\alpha/2$ in each tail, leaving central probability $1 - \alpha$.

Example 8.2 (Confidence interval for a proportion)
For $X_1, \ldots, X_n \sim \mathrm{Ber}(p)$, the CLT gives $\hat{p}_n \approx \mathcal{N}(p, \mathrm{se}^2)$ for large n, so an approximate $1 - \alpha$ interval for p is

$$\hat{p}_n \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_n(1-\hat{p}_n)}{n}}.$$

Because we substitute $\widehat{\mathrm{se}}$ for the unknown se, the coverage is only approximately correct, improving as n grows.

Example 8.3 (Known variance: a bottling machine)
A machine fills bottles with volumes that are $\mathcal{N}(\mu, \sigma^2)$ with $\sigma = 5$ ml *known*. A sample of 16 bottles has $\bar{x} = 743$ ml. Construct a 95% confidence interval for μ.

Solution. With σ known, $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly, so $\mathrm{se} = \sigma/\sqrt{n}$ and the interval is

$$\bar{x} \pm z_{.025}\frac{\sigma}{\sqrt{n}} = 743 \pm 1.96 \cdot \frac{5}{\sqrt{16}} = (740.55,\ 745.45)\ \text{ml}.$$

8.4 The t Distribution

In practice σ is rarely known and must be estimated from the data by the sample standard deviation S_n. For *small* samples this substitution is not negligible, and the right reference distribution is no longer the normal but the t distribution.

Definition 8.3. If $Z \sim \mathcal{N}(0,1)$ and χ^2_n (a *chi-square* variable with n degrees of freedom—the sum of squares of n independent standard normals) are independent, then $T_n = Z/\sqrt{\chi^2_n/n}$ has a *t distribution with n degrees of freedom.*

Like the normal, the t density is symmetric about zero, but it has *thicker tails*, reflecting the extra uncertainty from estimating σ (Figure 8.4). As the degrees of freedom grow, $\chi^2_n/n \to 1$ by the law of large numbers, and the t density approaches $\mathcal{N}(0,1)$ (Figure 8.3).

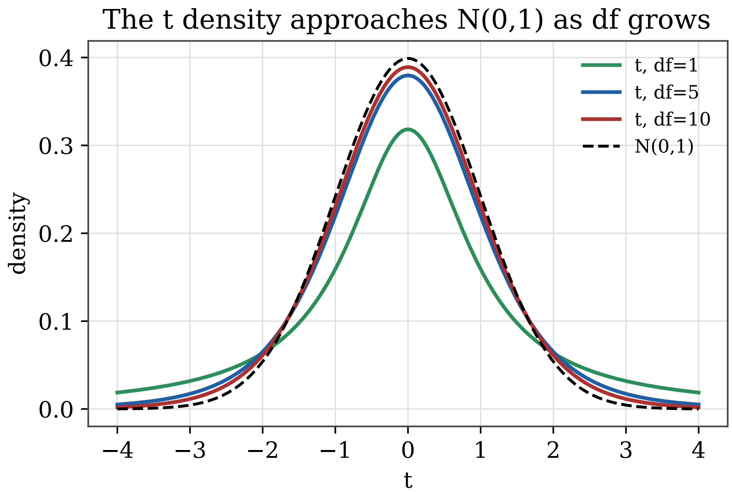


Figure 8.3. The t density with 1, 5, and 10 degrees of freedom, approaching $\mathcal{N}(0,1)$ (dashed) as the degrees of freedom increase.

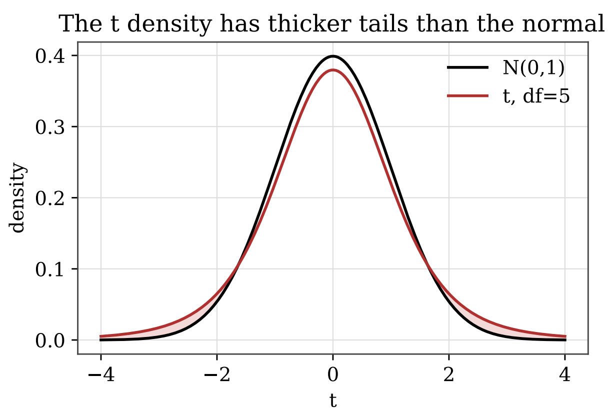


Figure 8.4. The t density (here with 5 df) has heavier tails than the standard normal, accounting for the added variability of an estimated σ.

The key fact is that for an i.i.d. $\mathcal{N}(\mu,\sigma^2)$ sample with σ unknown,

$$\frac{\sqrt{n}\,(\bar{X}_n - \mu)}{S_n} \sim t_{n-1},$$

exactly, not merely approximately. This yields an exact confidence interval $\bar{x} \pm t_{\alpha/2,\,n-1}\, s/\sqrt{n}$ for μ.

Example 8.4 (Unknown variance, small sample: caloric content of coal)
Twenty-two measurements of the gross caloric content of a coal shipment give $\bar{x} = 31.012$ MJ/kg and $s = 0.1294$, with the accuracy of the method in doubt (so σ is treated as unknown). Find a 95% confidence interval for the true mean μ.

Solution. With $n = 22$ we use $t_{.025,\,21} = 2.080$:

$$31.012 \pm 2.080 \cdot \frac{0.1294}{\sqrt{22}} = (30.954,\ 31.069).$$

This interval is about 50% wider than the corresponding known-σ interval would be, both because we use the larger estimate s in place of σ and because the critical value 2.080 exceeds 1.96. Estimating the spread costs us precision.

Example 8.5 (Large sample: t and z nearly agree)
A normal sample of size 34 has $\bar{x} = 3.54$ and $s = 0.13$. Construct a 98% confidence interval for μ.

Solution. The exact interval uses $t_{.01,\,33} \approx 2.45$:

$$3.54 \pm 2.45 \cdot \frac{0.13}{\sqrt{34}} \approx (3.49,\ 3.59),$$

while the large-sample approximation with $z_{.01} = 2.33$ gives a slightly narrower $(3.49, 3.59)$ as well. For samples this large the t and normal intervals are nearly identical, which is why the normal approximation is so widely used.

We summarize the standard intervals.

Situation	Target	Sampling law of $\bar{X}$	$1-\alpha$ interval
Ber(p)	p	$\approx$ normal (CLT)	$\bar{x} \pm z_{\alpha/2}\sqrt{\bar{x}(1-\bar{x})/n}$
$\mathcal{N}(\mu, \sigma^2)$, σ known	μ	exactly normal	$\bar{x} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$
$\mathcal{N}(\mu, \sigma^2)$, σ unknown	μ	t_{n-1}	$\bar{x} \pm t_{\alpha/2,\,n-1}\,s/\sqrt{n}$

8.5 Hypothesis Testing

The third inference problem starts from a default theory—the *null hypothesis* H_0—and asks whether the data provide enough evidence to reject it in favor of an *alternative* H_1. The logic mirrors a criminal trial: we presume H_0 (innocence) and reject only on strong evidence.

Two errors are possible. A *type I error* rejects a true H_0; its probability is denoted α (the same α as the significance level). A *type II error* retains H_0 when H_1 is true; its probability is denoted β.

	Retain H_0	Reject H_0
H_0 true	correct	type I error (α)
H_1 true	type II error (β)	correct

For example, to test whether a coin is fair we would set $H_0 : p = \frac{1}{2}$ against $H_1 : p \neq \frac{1}{2}$; to calibrate an automated speed trap so that at most 5% of ticketed drivers were not actually speeding, we would fix the type I error at 0.05. The machinery for carrying out such tests—test statistics, p-values, and significance levels—is developed in Chapters 11 and 12.

CHAPTER 9

Point Estimation: Unbiasedness and Efficiency

The previous chapter introduced point estimators and the notions of bias and variance. This chapter studies them in depth. We first build estimators from data by matching sample quantities to model quantities, then ask which estimators are *unbiased*, see how nonlinearity creates bias, and finally compare unbiased estimators by their variance to find the most *efficient*.

9.1 Estimators Built from Data

A natural way to estimate a parameter is to match a population quantity—a mean, a probability—to its sample counterpart. The law of large numbers is what makes this work.

Example 9.1 (A binomial parameter from queue data)

At a station, the number of women in each of 100 queues of length 10 is recorded. Modeling the count in a queue as $\mathrm{Bin}(10, p)$, estimate p.

Solution. If each of the 10 positions is independently a woman with probability p, the count is $\mathrm{Bin}(10, p)$ with mean $10p$. The sample mean of the counts is $\bar{x} = 4.35$, which we equate to $10p$, giving $\hat{p} = 0.435$. We matched the model's mean to the data's mean.

Example 9.2 (Two estimators for a geometric parameter)

The number of cycles to pregnancy for 93 women is recorded, modeled as $\mathrm{Geo}(p)$. Propose estimators of p.

Solution. Two natural choices arise. Since $\mathbb{E}[X] = 1/p$, we can estimate $\mathbb{E}[X]$ by the sample mean and invert: $\hat{p} = 1/\bar{x} = 93/331$. Alternatively, since $p = \mathbb{P}(X = 1)$, we can use the observed fraction of women who conceived in the first cycle, $\hat{p} = 29/93$. The two estimators—0.281 and 0.312—need not agree; choosing between them is exactly the kind of question this chapter addresses. (For the 474 nonsmoking women, the analogous estimates are $474/1285 = 0.369$ and $198/474 = 0.418$.)

Example 9.3 (Was London bombed at random? A Poisson fit)

An area of South London was divided into 576 quarter-kilometer squares, and the number of V-bomb hits in each was tallied:

hits	0	1	2	3	4	5	6	7
squares	229	211	93	35	7	0	0	1

If hits fall purely at random, the count per square should be Poisson. Estimate its mean and assess the fit.

Solution. Modeling the count as $\mathrm{Pois}(\mu)$ with $\mathbb{E}[X] = \mu$, we estimate μ by the average number of hits per square,

$$\hat{\mu} = \frac{0 \cdot 229 + 1 \cdot 211 + 2 \cdot 93 + \cdots + 7 \cdot 1}{576} \approx 0.93.$$

Plugging $\hat{\mu}$ into the Poisson pmf and comparing predicted to observed relative frequencies:

hits	0	1	2	3	4
observed	0.40	0.37	0.16	0.06	0.01
Poisson	0.39	0.37	0.17	0.05	0.01

The agreement is remarkable—evidence that the bombing was indeed random rather than precisely targeted. A formal version of this comparison, the chi-square goodness-of-fit test, appears in Chapter 12.

9.2 Unbiased Estimators

Recall that $\hat{\theta}_n$ is *unbiased* for θ if $\mathbb{E}[\hat{\theta}_n] = \theta$. Unbiasedness means the estimator is correct *on average*, with no systematic over- or under-shooting.

Example 9.4 (An unbiased estimator of θ^2)

Let $X_1, \ldots, X_n$ be a sample from the uniform distribution on $[-\theta, \theta]$. Show that $T = \frac{3}{n}\sum_{i=1}^n X_i^2$ is unbiased for θ^2.

Solution. For a $\mathrm{Unif}(-\theta, \theta)$ variable, $\mathbb{E}[X_i] = 0$ and $\mathrm{Var}(X_i) = \frac{(2\theta)^2}{12} = \frac{\theta^2}{3}$, so $\mathbb{E}[X_i^2] = \mathrm{Var}(X_i) + \mathbb{E}[X_i]^2 = \frac{\theta^2}{3}$. By linearity,

$$\mathbb{E}[T] = \frac{3}{n}\sum_{i=1}^n \mathbb{E}[X_i^2] = \frac{3}{n} \cdot n \cdot \frac{\theta^2}{3} = \theta^2.$$

Example 9.5 (An unbiased estimator of a probability)

For a $\mathrm{Geo}(p)$ sample, estimate $\mathbb{P}(X \leq 3)$ by the relative frequency $S = \frac{1}{n}\#\{i : X_i \leq 3\}$. Show S is unbiased.

Solution. Introduce indicators $Y_i = \mathbf{1}\{X_i \leq 3\}$, so $\mathbb{E}[Y_i] = \mathbb{P}(X_i \leq 3)$. Then

$$\mathbb{E}[S] = \frac{1}{n}\sum_{i=1}^n \mathbb{E}[Y_i] = \mathbb{P}(X \leq 3).$$

A relative frequency is always an unbiased estimator of the corresponding probability—no matter the underlying distribution.

Remark 9.1 (Why the sample variance divides by $n-1$)**.** We can now explain the divisor $n-1$ promised in Chapter 7. For an i.i.d. sample with variance σ^2, the estimator $\frac{1}{n-1}\sum(X_i - \bar{X}_n)^2$ is exactly unbiased for σ^2, whereas dividing by n would systematically underestimate it (because the deviations are taken about the sample mean, which is itself pulled toward the data). The correction factor $\frac{n}{n-1}$ removes this downward bias.

9.3 Bias from Nonlinearity

Unbiasedness is fragile under nonlinear transformation: even if T is unbiased for θ, $g(T)$ is usually *biased* for $g(\theta)$. Jensen's inequality (Chapter 6) tells us the *direction* of the bias.

Example 9.6 (A square root introduces bias)
With $T = \frac{3}{n}\sum X_i^2$ unbiased for θ^2 as above, is $\sqrt{T}$ unbiased for θ?

Solution. The function $g(x) = \sqrt{x}$ is concave, so by Jensen's inequality $\mathbb{E}[\sqrt{T}] \leq \sqrt{\mathbb{E}[T]} = \sqrt{\theta^2} = \theta$. Thus $\sqrt{T}$ has *negative* bias: it tends to underestimate θ.

Example 9.7 (Inverting the sample mean)
For a $\text{Geo}(p)$ sample, the law of large numbers suggests $T = 1/\bar{X}_n$ as an estimator of p (since $\mathbb{E}[X] = 1/p$). Is it unbiased?

Solution. The function $g(x) = 1/x$ is convex on $(0, \infty)$, so Jensen's inequality gives $\mathbb{E}[1/\bar{X}_n] > 1/\mathbb{E}[\bar{X}_n] = p$. Hence T has *positive* bias—it tends to overestimate p. Inverting an unbiased estimator does not preserve unbiasedness.

Example 9.8 (An exact bias calculation)
For a $\text{Pois}(\mu)$ sample, the probability of zero arrivals is $e^{-\mu}$, which one might estimate by $T = e^{-\bar{X}_n}$. Compute $\mathbb{E}[T]$.

Solution. Write $T = e^{-Z/n}$ with $Z = \sum X_i \sim \text{Pois}(n\mu)$. Then

$$\mathbb{E}[T] = \sum_{k=0}^{\infty} e^{-k/n}\,\frac{(n\mu)^k}{k!}e^{-n\mu} = e^{-n\mu}\sum_{k=0}^{\infty}\frac{(n\mu e^{-1/n})^k}{k!} = e^{-n\mu}\,e^{n\mu e^{-1/n}} = e^{-n\mu(1-e^{-1/n})}.$$

Since $1 - e^{-1/n} \neq 1/n$, this is not equal to $e^{-\mu}$: the estimator is biased, though the bias vanishes as $n \to \infty$.

9.4 Efficiency: Comparing Unbiased Estimators

Among unbiased estimators, the better one has the smaller variance (equivalently, since bias is zero, the smaller MSE). Such an estimator is more *efficient*.

Example 9.9 (Optimally combining two sample means)
Two independent samples, of sizes n and m from the same distribution with mean μ, give sample means $\bar{X}_n$ and $\bar{Y}_m$. Combine them as $T = r\bar{X}_n + (1-r)\bar{Y}_m$. (a) Show T is unbiased. (b) Find the r minimizing its variance.

Solution. (a) By linearity, $\mathbb{E}[T] = r\mu + (1-r)\mu = \mu$ for every r, so T is unbiased. (b) By independence,

$$\mathrm{Var}(T) = r^2\frac{\sigma^2}{n} + (1-r)^2\frac{\sigma^2}{m}.$$

Differentiating in r and setting the derivative to zero gives $\frac{2r}{n} - \frac{2(1-r)}{m} = 0$, i.e. $r = \frac{n}{n+m}$. The optimal weights are proportional to the sample sizes—the larger sample is trusted more.

Example 9.10 (Pooling two variance estimators)

Independent normal samples of sizes n and m have sample variances S_X^2 and S_Y^2, both unbiased for the common σ^2, with $\mathrm{Var}(S_X^2) = \frac{2\sigma^4}{n-1}$ and $\mathrm{Var}(S_Y^2) = \frac{2\sigma^4}{m-1}$. Among estimators $aS_X^2 + bS_Y^2$, find the best.

Solution. For unbiasedness, $\mathbb{E}[aS_X^2 + bS_Y^2] = (a+b)\sigma^2 = \sigma^2$ forces $a + b = 1$. Writing $b = 1 - a$,

$$\mathrm{Var}(aS_X^2 + (1-a)S_Y^2) = 2\sigma^4\Big(\frac{a^2}{n-1} + \frac{(1-a)^2}{m-1}\Big).$$

Minimizing over a gives $a = \frac{n-1}{n+m-2}$ and hence $b = \frac{m-1}{n+m-2}$. This is the *pooled* variance estimator, weighting each sample by its degrees of freedom—the standard estimator behind the two-sample t-test.

The maximum likelihood method of the next chapter provides a systematic recipe for constructing estimators that, under mild conditions, are asymptotically unbiased and asymptotically the most efficient of all.

CHAPTER 10

Maximum Likelihood Estimation

The estimators of the previous chapter were built one at a time, by ad hoc reasoning. We now meet a single, general recipe for producing good estimators—*maximum likelihood*. The principle is disarmingly simple: choose the parameter value that makes the observed data most probable.

10.1 The Maximum Likelihood Principle

Principle 10.1 (Maximum likelihood)**.** Given a dataset, estimate the unknown parameter by the value that makes the observed data as likely as possible.

A small story fixes the idea. A dealer is offered one of two batches of chips: one is about 10% defective, the other about 50%, but the seller has lost track of which is which. The dealer tests 10 chips from a batch and finds exactly one defective. Which batch is it more likely to be?

Example 10.1 (Which batch?)

Test 10 chips; let $R_i = 1$ if chip i is defective. With one defective among ten, the probability of the observed pattern is $p(1-p)^9$. Compare the two candidate values of p:

$$\text{(10\% batch)} \;\; (0.1)(0.9)^9 \approx 0.039, \qquad \text{(50\% batch)} \;\; (0.5)(0.5)^9 \approx 0.00098.$$

The observed data are about forty times more likely under the 10% batch, so the dealer should believe it is that one. This is maximum likelihood reasoning in miniature.

If instead the *first three* chips were defective, the data probability becomes $p^3(1-p)^7$, now favoring the 50% batch (≈ 0.00098 versus ≈ 0.00048): three early defectives are better explained by the high-defect batch. The data, not a prior hunch, decide.

Example 10.2 (Which die?)

A die D_1 has five red faces and one white; D_2 has one red and five white. One die is chosen and rolled until red appears, three separate times, giving first-red on the 3rd, 5th, and 4th rolls. Which die is it likely to be?

Solution. The rolls are independent $\mathrm{Geo}(p)$ with p the chance of red, so the likelihood is

$$L(p) = (1-p)^2 p \cdot (1-p)^4 p \cdot (1-p)^3 p = p^3(1-p)^9.$$

For D_1 ($p = \frac{5}{6}$): $L(\frac{5}{6}) \approx 5.74 \times 10^{-8}$. For D_2 ($p = \frac{1}{6}$): $L(\frac{1}{6}) \approx 8.97 \times 10^{-4}$. The data are vastly more likely under D_2, so it is almost surely the mostly-white die.

10.2 The Likelihood Function

For a sample $x_1, \ldots, x_n$ from a distribution with parameter θ, the *likelihood function* is the probability (discrete case) or density (continuous case) of the data, viewed as a function of θ:

$$L(\theta) = \prod_{i=1}^{n} p_\theta(x_i) \quad \text{(discrete)}, \qquad L(\theta) = \prod_{i=1}^{n} f_\theta(x_i) \quad \text{(continuous)}.$$

The *maximum likelihood estimate* (MLE) is the maximizer $\hat{\theta} = \arg\max_\theta L(\theta)$.

Example 10.3 (The exponential rate)

For a sample from $\text{Exp}(\lambda)$, find the MLE of λ.

Solution. The likelihood is

$$L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum x_i}.$$

Differentiating and setting $L'(\lambda) = 0$ gives $1 - \lambda \bar{x}_n = 0$, so the MLE is $\hat{\lambda} = 1/\bar{X}_n$. (One checks the second derivative is negative there, confirming a maximum.)

Example 10.4 (An endpoint that defies calculus)

A sample $x_1, x_2, x_3 = 0.98, 1.57, 0.31$ comes from $\text{Unif}(0, \theta)$ with θ unknown. Find the MLE of θ.

Solution. Each density is $1/\theta$ on $[0, \theta]$ and 0 outside, so the likelihood is

$$L(\theta) = \begin{cases} 1/\theta^3, & \theta \geq \max(x_i) = 1.57, \\ 0, & \theta < 1.57. \end{cases}$$

Differentiating would only show L is decreasing where it is positive; the maximum sits at the *boundary*. As Figure 10.1 shows, $L(\theta)$ jumps up to its largest value exactly at $\theta = 1.57$ and decays thereafter, so $\hat{\theta} = \max(x_1, x_2, x_3) = 1.57$. In general the MLE for the upper endpoint is $\max\{X_1, \ldots, X_n\}$.

10.3 The Loglikelihood

Because L is a product, differentiating it directly means repeated use of the product rule. Taking logarithms converts the product into a sum, which is far easier to differentiate; and since the logarithm is increasing, the *loglikelihood* $\ell(\theta) = \ln L(\theta)$ has its maximum at the same θ as L.

Example 10.5 (Cycles to pregnancy, with censoring)

For 100 smoking women, the number of cycles to pregnancy is modeled as $\text{Geo}(p)$; 7 women had an unknown number exceeding 12. Counting 29 ones, 16 twos, ..., and 7 values above 12, the likelihood works out (up to a constant) to

$$L(p) = C\, p^{93}(1 - p)^{322}.$$

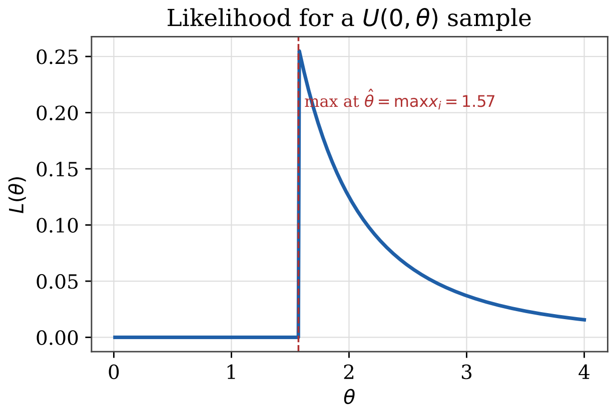


Figure 10.1. Likelihood for a $\mathrm{Unif}(0,\theta)$ sample. It is zero until θ reaches the largest observation, then decreases—so the maximum is at the boundary, where calculus alone would not find it.

Find $\hat{p}$.

Solution. The constant C counts arrangements and does not affect the maximizer. Differentiating,

$$L'(p) = C\,p^{92}(1-p)^{321}\big(93(1-p) - 322p\big) = C\,p^{92}(1-p)^{321}(93 - 415p),$$

which vanishes (in the interior) at $p = 93/415 \approx 0.224$. The loglikelihood $\ell(p) = \ln C + 93\ln p + 322\ln(1-p)$ has the same maximizer (Figure 10.2). Notably 0.224 is well below the naive estimate $29/100 = 0.29$ that ignores the censored women. For the 474 nonsmokers the likelihood is $\propto p^{474}(1-p)^{955}$, giving $\hat{p} = 474/1429 \approx 0.33$.

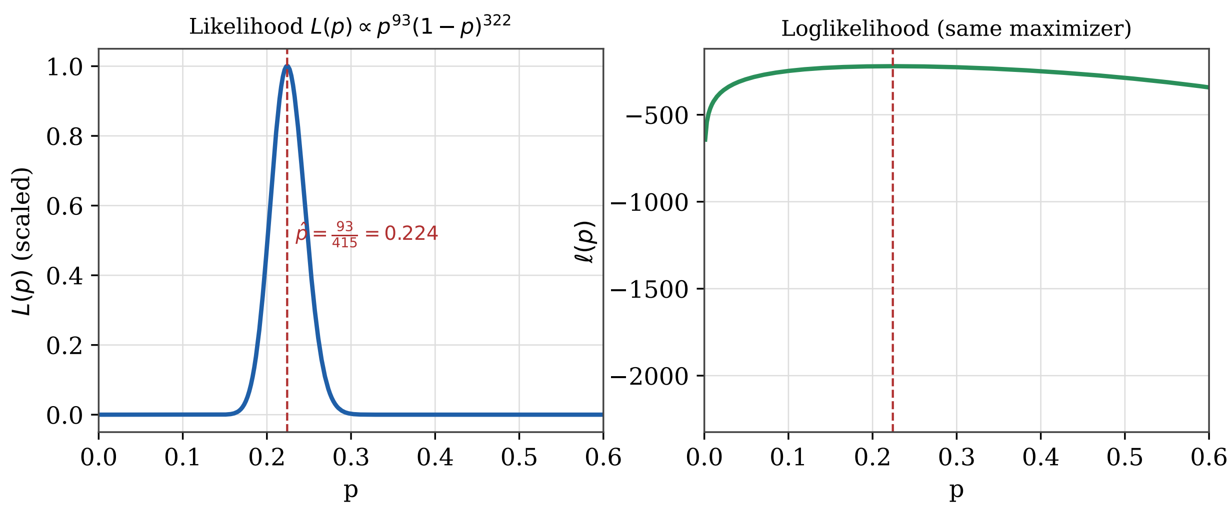


Figure 10.2. Likelihood $L(p) \propto p^{93}(1-p)^{322}$ (left) and its loglikelihood (right) for the smokers. Both peak at the same value $\hat{p} = 93/415 \approx 0.224$; the loglikelihood is easier to differentiate.

10.4 The Normal and Poisson MLEs

Example 10.6 (Estimating the mean and standard deviation of a normal)
For an i.i.d. $\mathcal{N}(\mu, \sigma^2)$ sample, find the MLEs of μ and σ.

Solution. The loglikelihood is

$$\ell(\mu, \sigma) = -n \ln \sigma - n \ln \sqrt{2\pi} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2.$$

Setting $\partial \ell / \partial \mu = 0$ gives $\hat{\mu} = \bar{x}_n$, and then $\partial \ell / \partial \sigma = 0$ gives

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x}_n)^2}.$$

(The same two answers follow if one parameter is estimated with the other held fixed—for instance, with μ known to be 0 the MLE of σ is $\sqrt{\frac{1}{n} \sum x_i^2}$.) Notice the MLE of the variance divides by n, not $n - 1$; we return to this below.

Example 10.7 (Poisson arrivals, and the probability of none)
Packets arrive at a server, modeled as $\text{Pois}(\mu)$ per minute. (a) Find the MLE of μ. (b) Find the MLE of the probability $e^{-\mu}$ of an idle minute.

Solution. (a) The likelihood is $L(\mu) = \prod_i \frac{\mu^{x_i}}{x_i!} e^{-\mu} = \frac{e^{-n\mu} \mu^{\sum x_i}}{\prod x_i!}$, with loglikelihood $\ell(\mu) = (\sum x_i) \ln \mu - n\mu - \ln \prod x_i!$. Then $\ell'(\mu) = \frac{\sum x_i}{\mu} - n = 0$ gives $\hat{\mu} = \bar{x}_n$, a maximum since $\ell'' < 0$. (Applied to the South London bomb data of Chapter 9, this gives $\hat{\mu} = 537/576 \approx 0.93$, confirming the earlier estimate as the MLE.) (b) By the invariance property below, the MLE of $e^{-\mu}$ is simply $e^{-\bar{x}_n}$.

10.5 Estimating a Population Size

Example 10.8 (The marked-bolt problem, revisited)
Recall the box of N bolts from Chapter 3. (a) If draws *with* replacement give a $\text{Geo}(1/N)$ sample, find the MLE of N. (b) If draws *without* replacement give a discrete uniform sample on $\{1, \ldots, N\}$, find the MLE of N.

Solution. (a) The loglikelihood for $\text{Geo}(1/N)$ data is $\ell(N) = -n \ln N + \left(\sum x_i - n\right) \ln(1 - \frac{1}{N})$; differentiating and solving $\ell'(N) = 0$ yields, after simplification, $\hat{N} = \bar{x}_n$. (b) For the discrete uniform, $L(N) = (1/N)^n$ when $N \geq \max x_i$ and 0 otherwise, a decreasing function of N on its support. The maximum is again at the boundary: $\hat{N} = \max\{X_1, \ldots, X_n\}$, the largest value seen—just as for the continuous uniform endpoint.

10.6 Properties of Maximum Likelihood Estimators

Beyond being a uniform recipe, MLEs enjoy several strong properties.

Invariance. If $\hat{\theta}$ is the MLE of θ and g is an invertible function, then $g(\hat{\theta})$ is the MLE of $g(\theta)$. (This is why $e^{-\bar{x}_n}$ is the MLE of $e^{-\mu}$, and why the MLE of σ^2 is the square of the MLE of σ.)

Asymptotic unbiasedness. An MLE may be biased in finite samples. The normal-variance MLE $D_n^2 = \frac{1}{n}\sum(X_i - \bar{X}_n)^2$ equals $\frac{n-1}{n}S_n^2$, so $\mathbb{E}[D_n^2] = \frac{n-1}{n}\sigma^2$—biased downward, but with the bias vanishing as $n \to \infty$. Under mild conditions every MLE is asymptotically unbiased: $\lim_{n\to\infty}\mathbb{E}[T_n] = \theta$.

Asymptotic minimum variance. The variance of any unbiased estimator is bounded below by the *Cramér–Rao* bound, and MLEs attain this bound asymptotically: no estimator does better in large samples.

Asymptotic normality. The standardized MLE $(\hat{\theta} - \theta)/\widehat{\mathrm{se}}$ converges to $\mathcal{N}(0,1)$. Consequently

$$\hat{\theta}_n \pm 1.96\,\widehat{\mathrm{se}}$$

is an approximate 95% confidence interval for θ.

Example 10.9 (Confidence intervals for the pregnancy parameters)
For the $\mathrm{Geo}(p)$ model, the MLE $\hat{p}$ has standard error $\mathrm{se} = \sqrt{p^2(1-p)/n}$. Using asymptotic normality, build approximate 95% intervals for the smokers ($\hat{p} = 0.224$, $n = 100$) and nonsmokers ($\hat{p} = 0.33$, $n = 486$).

Solution. The interval is $\hat{p} \pm 1.96\sqrt{\hat{p}^2(1-\hat{p})/n}$. For the smokers this gives roughly $(0.19, 0.26)$, and for the nonsmokers roughly $(0.31, 0.35)$. The two intervals do not overlap, strong evidence that smokers and nonsmokers really do have different per-cycle conception probabilities.

CHAPTER 11

The Method of Least Squares

When two quantities are related—a wood's density and its hardness, calcium in drinking water and a town's mortality—we often want to summarize the relationship by a line and to use it for prediction. The method of *least squares* fits such a line by minimizing the total squared vertical distance between the data and the line. Remarkably, this turns out to be maximum likelihood in disguise, under the assumption that the measurement errors are normal.

11.1 The Simple Linear Regression Model

The *simple linear regression model* for a bivariate dataset $(x_1, y_1), \ldots, (x_n, y_n)$ treats the x_i as fixed and the responses as random:

$$Y_i = \alpha + \beta x_i + U_i, \qquad i = 1, \ldots, n,$$

where $U_1, \ldots, U_n$ are independent with mean 0 and variance σ^2. Equivalently, $Y_i \sim \mathcal{N}(\alpha + \beta x_i, \sigma^2)$: at each setting of x there is a whole population of possible responses, normally distributed about the line $\mathbb{E}[Y] = \alpha + \beta x$, all sharing the same variance. The line's height shifts with x, but the spread does not.

11.2 Least Squares Is Maximum Likelihood

To fit the line we choose α, β minimizing the sum of squared vertical distances

$$S(\alpha, \beta) = \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2.$$

This is not an arbitrary choice. Under the normal model the loglikelihood is

$$\ell(\alpha, \beta, \sigma) = -n \ln \sigma - n \ln \sqrt{2\pi} - \frac{1}{2\sigma^2} \sum_{i=1}^{n} (y_i - \alpha - \beta x_i)^2,$$

which, for any fixed σ, is maximized precisely when $S(\alpha, \beta)$ is minimized. Least squares *is* maximum likelihood for normal errors—one reason normal-error models dominate applications. (Differentiating ℓ in σ also gives the MLE $\hat{\sigma} = \sqrt{\frac{1}{n} \sum (Y_i - \hat{\alpha} - \hat{\beta} x_i)^2}$.)

As Figure 11.1 suggests, $S(\alpha, \beta)$ is an upward-opening bowl (an elliptic paraboloid), so it has a unique minimum—except in the degenerate case where all x_i are equal.

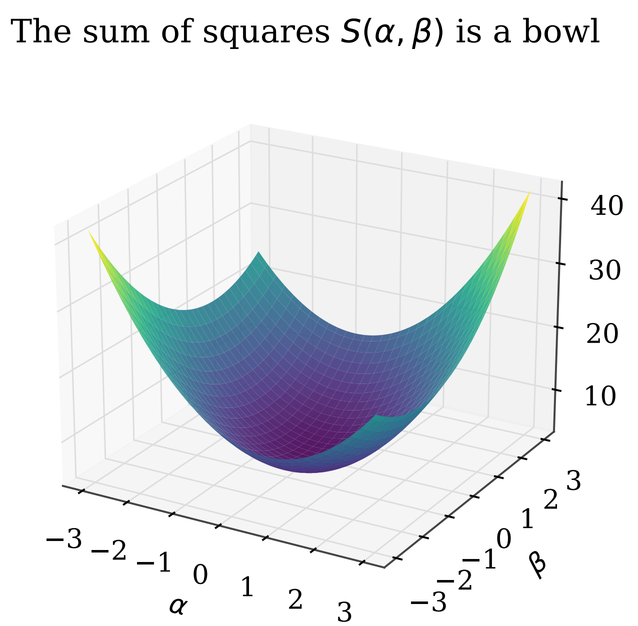


Figure 11.1. The sum-of-squares surface $S(\alpha, \beta)$ is a bowl with a single lowest point $(\hat{\alpha}, \hat{\beta})$.

11.3 The Normal Equations and the Estimates

Setting the partial derivatives of S to zero yields the *normal equations*

$$n\alpha + \beta \sum x_i = \sum y_i, \qquad \alpha \sum x_i + \beta \sum x_i^2 = \sum x_i y_i,$$

a 2×2 linear system whose solution is the pair of least-squares estimates

$$\hat{\beta} = \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}, \qquad \hat{\alpha} = \bar{y}_n - \hat{\beta}\, \bar{x}_n.$$

The second formula shows that the fitted line always passes through the center of mass $(\bar{x}_n, \bar{y}_n)$ of the data.

Example 11.1 (A three-point fit)

Fit the least-squares line to $(1, 2), (3, 1.8), (5, 1)$, and find the residuals.

Solution. With $\sum x_i = 9$, $\sum y_i = 4.8$, $\sum x_i^2 = 35$, $\sum x_i y_i = 12.4$, and $n = 3$,

$$\hat{\beta} = \frac{3(12.4) - 9(4.8)}{3(35) - 9^2} = -0.25, \qquad \hat{\alpha} = 1.6 - (-0.25)(3) = 2.35,$$

so the line is $y = 2.35 - 0.25x$ (Figure 11.2). The residuals $r_i = y_i - \hat{\alpha} - \hat{\beta} x_i$ are -0.1, 0.2, -0.1, which sum to zero—as they always must.

Example 11.2 (Predicting timber hardness)

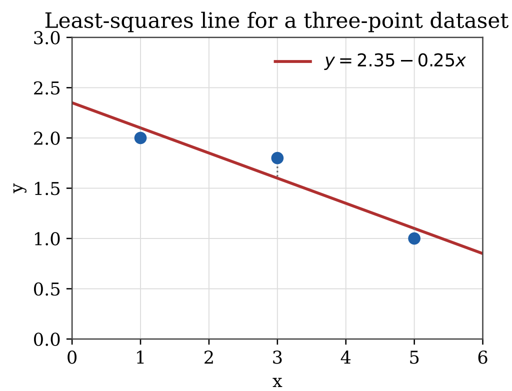


Figure 11.2. The least-squares line for three points; the dotted segments are the residuals being minimized.

For 36 Australian hardwoods, regressing Janka hardness on density gives $\hat{\alpha} = -1160.5$ and $\hat{\beta} = 57.51$. Predict the hardness of a specimen of density 65.

Solution. Substitute into the fitted line: $\hat{y} = -1160.5 + 57.51(65) \approx 2577.7$. The regression line turns an easily-measured density into a prediction of the hard-to-measure hardness.

11.4 Unbiasedness of the Estimators

The least-squares estimators are unbiased.

Example 11.3 (**$\hat{\beta}$ is unbiased**)

Show $\mathbb{E}[\hat{\beta}] = \beta$ in the model $Y_i = \alpha + \beta x_i + U_i$.

Solution. Since the x_i are constants and $\mathbb{E}[Y_i] = \alpha + \beta x_i$, linearity gives

$$\mathbb{E}[\hat{\beta}] = \frac{n\sum x_i \mathbb{E}[Y_i] - (\sum x_i)(\sum \mathbb{E}[Y_i])}{n\sum x_i^2 - (\sum x_i)^2} = \frac{n\sum x_i(\alpha + \beta x_i) - (\sum x_i)\sum(\alpha + \beta x_i)}{n\sum x_i^2 - (\sum x_i)^2}.$$

The α-terms cancel and the numerator collapses to $\beta(n\sum x_i^2 - (\sum x_i)^2)$, which is exactly the denominator times β. Hence $\mathbb{E}[\hat{\beta}] = \beta$. It follows that $\mathbb{E}[\hat{\alpha}] = \mathbb{E}[\bar{Y}_n] - \bar{x}_n\mathbb{E}[\hat{\beta}] = (\alpha + \beta\bar{x}_n) - \bar{x}_n\beta = \alpha$, so $\hat{\alpha}$ is unbiased too.

For the error variance, the MLE $\frac{1}{n}\sum(Y_i - \hat{\alpha} - \hat{\beta}x_i)^2$ turns out to have expectation $\frac{n-2}{n}\sigma^2$, so the *unbiased* estimator divides by $n - 2$ instead:

$$\hat{\sigma}^2 = \frac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{\alpha} - \hat{\beta}x_i)^2.$$

We lose two degrees of freedom because two parameters, α and β, were estimated from the same data.

11.5 Residuals and Model Checking

The *residuals* $r_i = y_i - \hat{\alpha} - \hat{\beta}x_i$ are the leftover vertical distances. They always sum to zero (because the line passes through $(\bar{x}, \bar{y})$), and plotting them against x_i is the standard diagnostic. If the linear model is right, the residual plot should look like featureless random scatter about zero, as for the black cherry tree data (Figure 11.3).

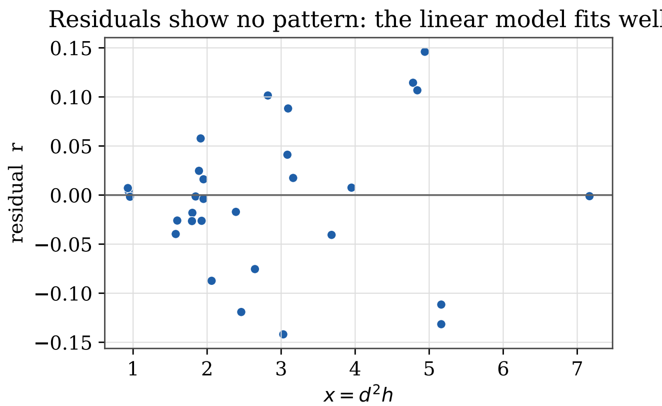


Figure 11.3. Residuals with no trend: the linear model is adequate.

A *pattern* in the residuals signals a misspecified model. For the timber data, the residuals from a straight-line fit bend in a parabola (Figure 11.4, left), suggesting a quadratic term:

$$Y_i = \alpha + \beta x_i + \gamma x_i^2 + U_i.$$

(This is still called a *linear* model—linear in the parameters α, β, γ, even though the fitted curve is a parabola; see Figure 11.5.) After adding the quadratic term, the residuals lose their bend but begin to *fan out* (Figure 11.4, right): their spread grows with x. This is *heteroscedasticity*—a violation of the equal-variance assumption—as opposed to the constant-variance ideal of *homoscedasticity.* One remedy is weighted least squares, which we do not pursue here.

11.6 Regression Through the Origin

Sometimes the intercept should be zero on physical grounds. For the black cherry tree data, the response y is timber volume and the predictor is $x = d^2h$ (diameter squared times height); as a tree shrinks to nothing, $x \to 0$ forces $y \to 0$, so the model $Y_i = \beta x_i + U_i$ with no intercept is natural.

Example 11.4 (The no-intercept slope)

Minimize $S(\beta) = \sum(y_i - \beta x_i)^2$ to derive the slope estimate.

Solution. Differentiating, $S'(\beta) = -2\sum(y_i - \beta x_i)x_i = 0$ gives

$$\hat{\beta} = \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i=1}^{n} x_i^2}.$$

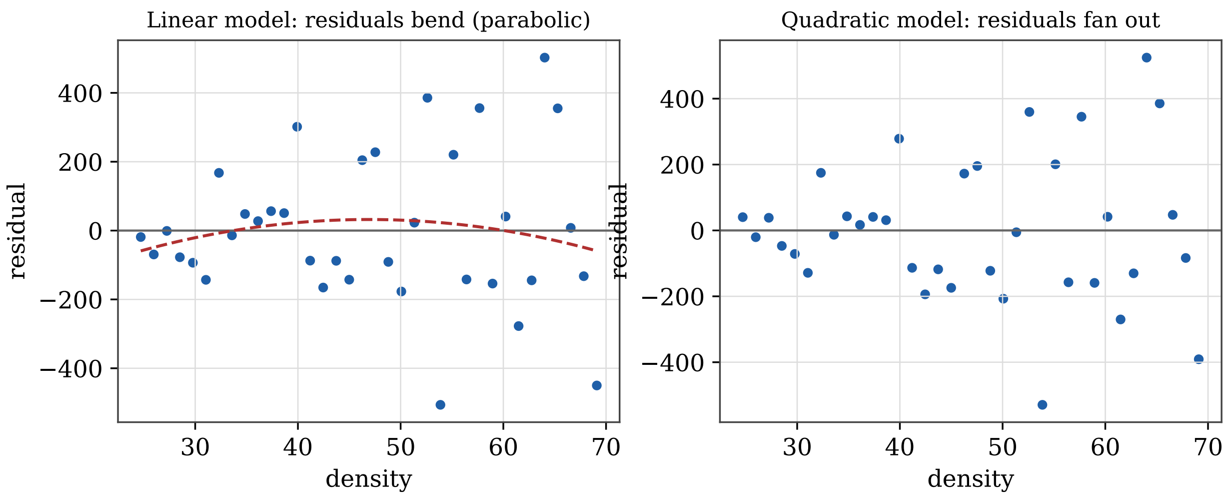


Figure 11.4. Left: a straight-line fit to the timber data leaves a parabolic trend. Right: after adding a quadratic term the trend is gone, but the residuals fan out, revealing heteroscedasticity.

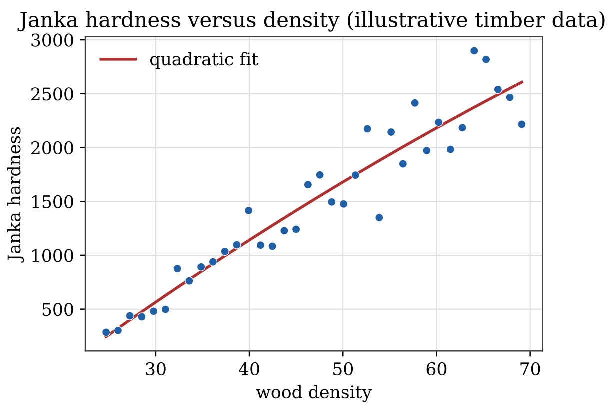


Figure 11.5. Janka hardness versus density with a fitted quadratic. The model is linear in its parameters even though the curve is not a straight line.

Example 11.5 (Three competing slope estimators)

For the no-intercept model, three natural estimators of β are the average of ratios $B_1 = \frac{1}{n}\sum \frac{Y_i}{x_i}$, the ratio of averages $B_2 = \frac{\sum Y_i}{\sum x_i}$, and the least-squares estimator $B_3 = \frac{\sum x_i Y_i}{\sum x_i^2}$. Show all three are unbiased, and compare them on the cherry tree data.

Solution. Each is a linear combination of the Y_i with $\mathbb{E}[Y_i] = \beta x_i$, so $\mathbb{E}[B_1] = \mathbb{E}[B_2] = \mathbb{E}[B_3] = \beta$: all unbiased. Numerically, using $\sum x_i = 87.456$, $\sum y_i = 26.486$, $\sum y_i/x_i = 9.369$, $\sum x_i y_i = 95.498$, $\sum x_i^2 = 314.644$, the three estimates are

$$B_1 \approx 0.302, \qquad B_2 \approx 0.303, \qquad B_3 \approx 0.3035.$$

They are close but not identical (Figure 11.6). Unbiasedness alone does not single out a winner—efficiency does, and the least-squares estimator B_3 is the one with smallest variance.

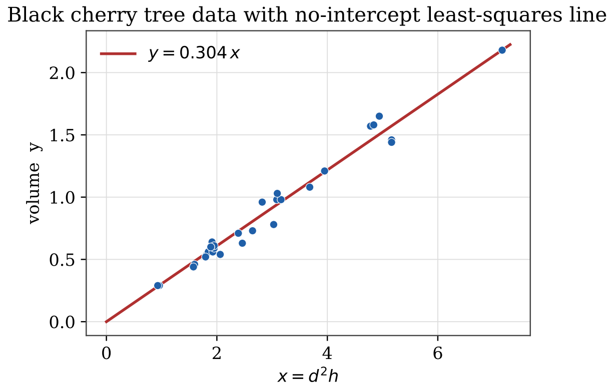


Figure 11.6. Black cherry tree volumes against $x = d^2h$ with the no-intercept least-squares line.

11.7 Testing Whether a Coefficient Belongs

A central question in regression is whether a coefficient is really nonzero, or whether the apparent effect could be chance. We test, say, $H_0 : \beta = \beta_0$ using a *test statistic*

$$T_b = \frac{\hat{\beta} - \beta_0}{S_b}, \qquad S_b^2 = \frac{n}{n\sum x_i^2 - (\sum x_i)^2}\,\hat{\sigma}^2,$$

which standardizes the estimate by an estimate of its standard deviation. Under H_0, T_b has a t distribution with $n-2$ degrees of freedom. Values near zero support H_0; large values argue against it.

The strength of evidence is summarized by a *p-value*—the probability, *if* H_0 *were true*, of a test statistic at least as extreme as the one observed. A small p-value is strong evidence against H_0. If we reject when the p-value falls below a threshold, that threshold is the *significance level*. A common informal scale:

p-value	evidence against H_0
< 0.01	very strong
0.01–0.05	strong
0.05–0.10	weak
> 0.10	little to none

Remark 11.1. The p-value is *not* $\mathbb{P}(H_0 \text{ true} \mid \text{data})$—that quantity belongs to Bayesian inference. The p-value is the smallest significance level at which the data would lead us to reject H_0.

Example 11.6 (Does the cherry-tree intercept belong?)
Fitting the cherry tree data *with* an intercept and testing $H_0 : \alpha = 0$ (we expect no intercept, since zero size should mean zero volume), the test statistic is $t_a = -0.428$ on 29 degrees of freedom, giving a one-sided p-value $\mathbb{P}(t_{29} \le -0.428) = 0.336$.

Solution. This p-value is large, so we retain H_0: the data are entirely consistent with a zero intercept, confirming the physical reasoning behind the no-intercept model.

Example 11.7 (Does calcium affect mortality?)

For 61 English towns, mortality rate y is regressed on calcium concentration x (Figure 11.7). We test $H_0 : \beta = 0$ against $H_1 : \beta < 0$ (more calcium, lower mortality). The fit gives $\hat{\beta} = -3.226$ and $s_b = 0.485$, so $t_b = -6.656$ on 59 degrees of freedom.

Solution. The corresponding p-value is about 5×10^{-9}—overwhelming evidence against H_0. We conclude that higher calcium concentrations are genuinely associated with lower mortality, not an artifact of chance.

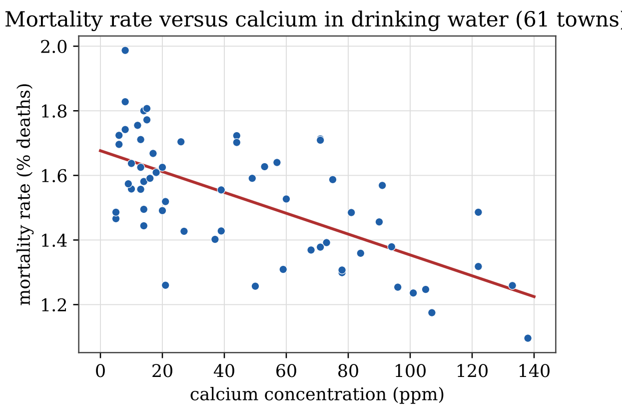


Figure 11.7. Mortality rate against calcium concentration for 61 towns, with the least-squares line. The downward slope is highly significant.

11.8 Reading a Regression Summary

Statistical software reports several diagnostics for a fitted regression. The most important is R^2, the *coefficient of determination*,

$$R^2 = 1 - \frac{SS_{\text{res}}}{SS_{\text{tot}}}, \qquad SS_{\text{res}} = \sum_i (y_i - \hat{y}_i)^2, \quad SS_{\text{tot}} = \sum_i (y_i - \bar{y})^2,$$

the fraction of the response's variation explained by the model. In simple linear regression R^2 is just the square of the empirical correlation coefficient. Figure 11.8 illustrates the idea: SS_{tot} measures spread about the flat mean line, SS_{res} about the fitted line; the more the line shrinks the squares, the closer R^2 is to one.

A summary also reports the *adjusted* R^2, which penalizes extra parameters and so guards against overfitting; the *residual standard error*, $\hat{\sigma}$ on $n-2$ degrees of freedom (two parameters estimated); and an *F-test* of whether the regression coefficients are jointly zero. In simple linear regression the F-test merely echoes the t-test for the slope, but it becomes genuinely informative when several predictors are present.

Finally, 95% confidence intervals for the parameters are obtained from the t distribution with $n-2$ degrees of freedom. For the mortality data, the interval for the calcium slope β is about $(-0.0042, -0.0023)$—comfortably below zero, consistent with the highly significant test above. For the quadratic timber fit, the intercept's interval $(-804, 556)$ comfortably contains zero, as physical reasoning predicts, while the quadratic coefficient's interval $(0.19, 0.82)$ excludes zero, confirming

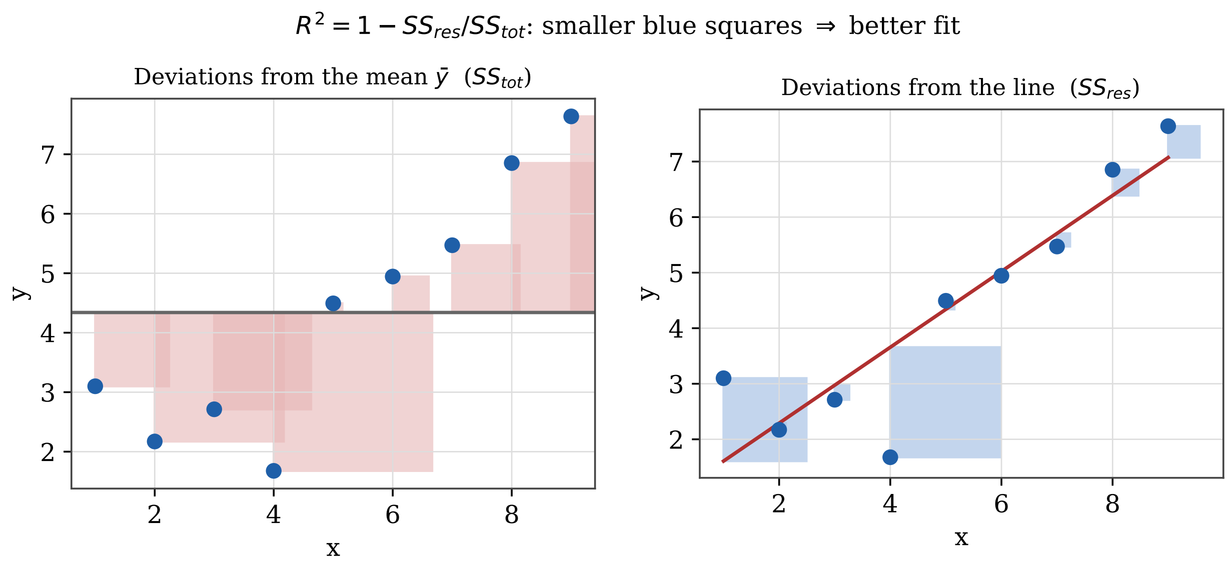


Figure 11.8. $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$. The red squares (left) measure deviations from the mean; the blue squares (right) measure deviations from the fitted line. Smaller blue squares mean a better fit and larger R^2.

that the curvature is real. Wide intervals like the timber ones can be narrowed only by gathering more data.

CHAPTER 12

Testing for Goodness of Fit

We close with a question that has run quietly beneath the whole second half of the book: does a proposed probability model actually *fit* the data? When we modeled bomb hits as Poisson or pea types by Mendel's ratios, we compared observed and predicted frequencies informally. This chapter makes the comparison rigorous through the likelihood ratio test and Pearson's chi-square test.

12.1 Is There a "Best" Test?

First, some vocabulary. A hypothesis fixing a parameter exactly, like $\theta = \theta_0$, is *simple*; one allowing a range, like $\theta > \theta_0$, is *composite*. A test of $H_0 : \theta = \theta_0$ against $H_1 : \theta \neq \theta_0$ is *two-sided*; a test against $H_1 : \theta > \theta_0$ is *one-sided*. Recall the two error probabilities, α (type I) and β (type II); the *power* of a test is $1 - \beta$, its chance of correctly rejecting a false H_0. The usual strategy fixes α and seeks the largest possible power. A test achieving the most power among all level-α tests is *most powerful*.

Most powerful tests are hard to find and often do not exist. But in the simplest case they do, and they take a memorable form.

Theorem 12.1 (Neyman–Pearson lemma)**.** *To test the simple $H_0 : \theta = \theta_0$ against the simple $H_1 : \theta = \theta_1$, form the likelihood ratio*

$$T = \frac{L(\theta_1)}{L(\theta_0)} = \frac{\prod_i f(x_i; \theta_1)}{\prod_i f(x_i; \theta_0)},$$

and reject H_0 when $T > k$. Choosing k so that $\mathbb{P}_{\theta_0}(T > k) = \alpha$ yields the most powerful level-α test.

In words: when both hypotheses are simple and there are no nuisance parameters, the optimal test compares how well the two parameter values explain the data, through a ratio of likelihoods.

12.2 The Likelihood Ratio Test

Real problems usually involve composite hypotheses or nuisance parameters, where the Neyman–Pearson lemma does not directly apply. The *likelihood ratio test* generalizes its idea. To test $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \notin \Theta_0$, the test statistic is

$$\lambda = 2 \ln\left(\frac{\max_{\theta \in \Theta} L(\theta)}{\max_{\theta \in \Theta_0} L(\theta)} \right) = 2 \ln\left(\frac{L(\hat{\theta})}{L(\hat{\theta}_0)} \right),$$

the (twice log) ratio of the best fit overall to the best fit allowed under H_0. Large λ means H_0 explains the data far worse than the unrestricted model, and counts against H_0. Its calibration is a theorem.

Theorem 12.2. *Under H_0, as $n \to \infty$ the statistic λ converges to a chi-square distribution whose degrees of freedom equal the difference in dimension between Θ and Θ_0. The p-value is the upper-tail probability $\mathbb{P}(\chi^2_{r-q} > \lambda)$.*

The chi-square family (Figure 12.1) thus governs goodness-of-fit testing, just as the normal and t families governed estimation.

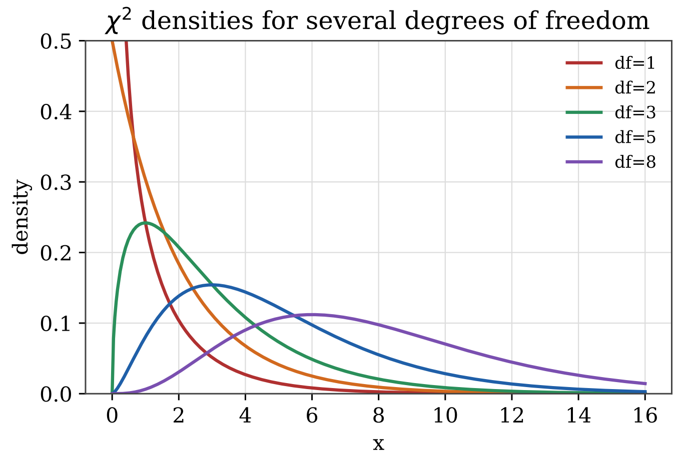


Figure 12.1. Chi-square densities. The degrees of freedom—the difference in dimension between the full and restricted parameter spaces—determine which curve calibrates the test.

12.3 The Multinomial Distribution

Goodness-of-fit data are counts in categories, which follow a *multinomial* distribution—the multivariate cousin of the binomial. Drawing n times independently from k categories with probabilities $p = (p_1, \dots, p_k)$, and letting X_j count category j, gives $X \sim \text{Mult}(n, p)$ with

$$\mathbb{P}(X = x) = \binom{n}{x_1 \cdots x_k} p_1^{x_1} \cdots p_k^{x_k}.$$

Each X_j is marginally $\text{Bin}(n, p_j)$, so $\mathbb{E}[X_j] = np_j$.

Example 12.1 (The multinomial MLE)

Show that the MLE of p for multinomial data is $\hat{p}_j = X_j/n$.

Solution. Ignoring a constant, the loglikelihood is $\ell(p) = \sum_j X_j \ln p_j$, to be maximized subject to $\sum_j p_j = 1$. Using a Lagrange multiplier λ, maximize $\sum_j X_j \ln p_j + \lambda(\sum_j p_j - 1)$. Setting $\partial/\partial p_j = \frac{X_j}{p_j} + \lambda = 0$ gives $\hat{p}_j = -X_j/\lambda$, and the constraint $\sum_j \hat{p}_j = 1$ forces $\lambda = -n$, so $\hat{p}_j = X_j/n$—each category's observed proportion.

12.4 Pearson's Chi-Square Test

To test a fully specified $H_0 : p = p_0$ against $H_1 : p \neq p_0$, Pearson's chi-square statistic compares observed counts X_j to the counts $E_j = np_j$ expected under H_0:

$$T = \sum_{j=1}^{k} \frac{(X_j - E_j)^2}{E_j}.$$

Under H_0, T converges to χ^2_{k-1} (one degree of freedom is lost to the constraint $\sum X_j = n$), and the p-value is $\mathbb{P}(\chi^2_{k-1} > t)$. Pearson's statistic is in fact a second-order Taylor approximation to the likelihood ratio statistic λ—a computational shortcut from 1900, when logarithms were laborious—so the two give nearly identical answers.

Example 12.2 (Mendel's peas)

Mendel's theory predicts that round-yellow, wrinkled-yellow, round-green, and wrinkled-green peas occur in proportions $p_0 = (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$. In $n = 556$ plants he observed $X = (315, 101, 108, 32)$. Test $H_0 : p = p_0$.

Solution. The expected counts are $np_0 = (312.75, 104.25, 104.25, 34.75)$ (Figure 12.2). Pearson's statistic is

$$t = \frac{(315 - 312.75)^2}{312.75} + \frac{(101 - 104.25)^2}{104.25} + \frac{(108 - 104.25)^2}{104.25} + \frac{(32 - 34.75)^2}{34.75} = 0.47.$$

With $k = 4$ categories there are $k - 1 = 3$ degrees of freedom, so the p-value is $\mathbb{P}(\chi^2_3 > 0.47) = 0.93$—no evidence against the theory. The likelihood ratio statistic gives almost the same value, $\lambda = 0.48$, with p-value 0.92. By either test, Mendel's data are strikingly consistent with his $9 : 3 : 3 : 1$ law.

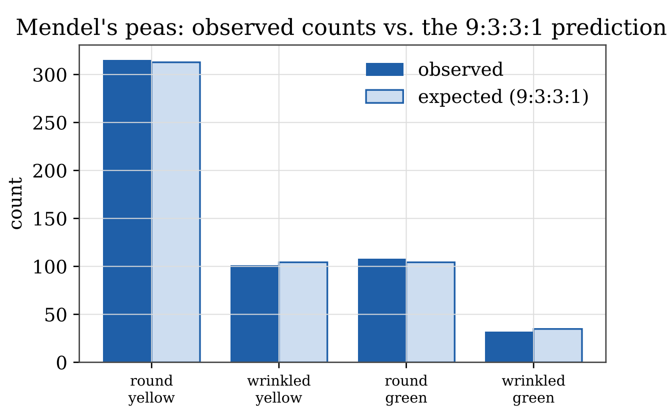


Figure 12.2. Mendel's observed pea counts against the $9 : 3 : 3 : 1$ prediction. The near-perfect agreement is reflected in a tiny chi-square statistic.

12.5 Goodness of Fit with Estimated Parameters

Often the model under test is not fully specified—we hypothesize a Poisson distribution but must estimate its mean from the data. Each parameter estimated costs one further degree of freedom.

Example 12.3 (Are accidents Poisson?)

The number of accidents per week at an intersection over $n = 50$ weeks was:

x	0	1	2	≥ 3
frequency	32	12	6	0

Test whether the weekly count is Poisson.

Solution. Under H_0 the count is $\text{Pois}(\mu)$ with μ unknown. Its MLE is the total accidents over total weeks, $\hat{\mu} = 24/50 = 0.48$. Pooling into the three categories $X = 0$, $X = 1$, $X \geq 2$, the estimated probabilities are

$$\hat{p}_1 = e^{-0.48} = 0.619, \quad \hat{p}_2 = 0.48e^{-0.48} = 0.297, \quad \hat{p}_3 = 1 - \hat{p}_1 - \hat{p}_2 = 0.084,$$

giving expected counts $E_1 = 30.95$, $E_2 = 14.85$, $E_3 = 4.20$. Then

$$t = \frac{(32 - 30.95)^2}{30.95} + \frac{(12 - 14.85)^2}{14.85} + \frac{(6 - 4.20)^2}{4.20} = 1.354.$$

With $k = 3$ categories we lose one degree of freedom to the constraint $\sum X_j = n$ and *another* to estimating μ, leaving $k - 2 = 1$. The p-value is $\mathbb{P}(\chi^2_1 > 1.354) = 0.24$, so we do not reject H_0: the data are consistent with a Poisson model for the accident counts.

This brings our tour full circle. We began by describing data, built the probability theory to model it, and developed the inferential tools—estimation, regression, and testing—to draw conclusions from it. Goodness-of-fit testing closes the loop, letting us ask of any model the most basic question of all: does it fit?